

**PRESSURE ULCER CLASSIFICATION OF STAGES I TO IV
USING DDPM, STYLEGAN2-ADA AND CNN**

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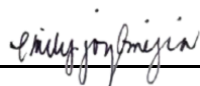
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RECOMMENDATION SHEET

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Chapter 1

INTRODUCTION

This research explores a hybrid deep learning pipeline that integrates Denoising Diffusion Probabilistic Models (DDPM) and StyleGAN2-ADA for data generation with Convolutional Neural Networks (CNNs), specifically DenseNet and YOLOv11 for automated classification of pressure ulcer stages (I–IV). By directly addressing inconsistent image quality across heterogeneous sources, the study targets improved robustness and accuracy under real-world, diverse, and uncontrolled imaging conditions. Empirically, the approach is evaluated across eight experimental runs that systematically combine data conditions (NON-AUGMENTED, DDPM, StyleGAN2-ADA, and DDPM+StyleGAN2-ADA) with model families (DenseNet and YOLOv11). This chapter presents the foundational aspects of the study and is organized into the following sections: (1) Context of the Study, (2) The Problem, Gap, or Opportunity, (3) Research Questions, (4) Research Objectives, (5) Significance of the Study, and (6) Scope and Limitations.

1.1 Context of the Study

Deep learning has been a significant advancement in the field of computer science in recent years, showing remarkable success in solving complex problems such as image classification and pattern recognition [1]. Its layered architecture, capable of automatically extracting hierarchical features from non-augmented data, has made it particularly valuable in domains requiring high precision and adaptability [1] [2]. One of the most revolutionary applications of deep learning has been in the field of medicine, where it has significantly enhanced diagnostic accuracy and decision-making efficiency [3]. In the classification and assessment of pressure ulcer stages, a common condition among elderly and immobile patients, deep learning models have shown promising

results in assisting medical professionals with more consistent and efficient evaluations of wound severity [4]. In addition to addressing the challenges of manual diagnosis, such as subjectivity and limited scalability, the integration of deep learning into healthcare offers a pathway to more personalized and accessible patient care.

Various deep learning models have emerged and been introduced in the industry with different strengths and purposes. Among the deep learning advancements, models like Denoising Diffusion Probabilistic Model (DDPM), StyleGAN2-ADA (Generative Adversarial Networks with Adaptive Discriminator Augmentation), and Convolutional Neural Networks (CNN) have shown promising performances in terms of realistic image generation, low-quality enhancement, object detection, and classification. The abilities of these models in terms of medical diagnosis have provided satisfactory results as they can deliver quick execution in terms of decision-making, timely support, and efficiency [5].

The diffusion model, Denoising Diffusion Probabilistic Model (DDPM), specializes in progressively applying Gaussian noise to an image and then reversing the process to obtain a high-quality version of the image [6]. Following this, StyleGAN2 (with ADA extension) generates realistic artificial images and discriminates against the result based on how ‘realistic’ the synthetic images are, while ADA allows the model to augment and learn from a small set of medical data [7]. Integrating DDPM with StyleGAN2 provides promising results, as stopping the process of the diffusion model halfway and implementing StyleGAN2 produces not only high-quality realistic images but also an increase in performance with no additional computation cost [8]. In this study, Early Stopped DDPM (for enhancement) and StyleGAN2-ADA (for augmentation) are treated as a preprocessing condition applied to the training set. The

comparison is factorial: each model is trained with and without this preprocessing to isolate preprocessing effects from model effects.

Convolutional Neural Networks (CNNs), often used for image-related tasks, work in layers to adaptively learn and classify the input image, allowing the model to recognize patterns [9]. The potential capabilities of these deep learning models regarding automatically diagnosing low-quality images with limited sets of data can establish a foundation for a broader application in the image processing pipeline. In classifying pressure ulcer stages, we compare two CNN-based models (DenseNet and YOLOv11) across four training data conditions: NON-AUGMENTED, DDPM, StyleGAN2-ADA, and DDPM+StyleGAN2-ADA. Validation and test sets remain NON-AUGMENTED. This 2×4 design estimates the main effects of model family and data condition, as well as their interaction, to assess whether generative preprocessing improves performance on mixed-quality clinical images.

The work by Lei et al. [9], titled “Convolutional Neural Network Models for Visual Classification of Pressure Ulcer Stages: Cross-Sectional Study”, evaluated different CNN architectures to classify pressure injuries in various stages and found promising performance with DenseNet, obtaining an accuracy of 93.71%. Dense Convolutional Network (DenseNet) is a model with exceptional performance compared to traditional convolutional networks, as various advancements were implemented to tackle limitations from other CNN architectures. This model works by leveraging dense connectivity and applying feature maps to every layer, improving the flow of the data throughout the network, and producing decent quality performances while needing fewer parameters and computation [11]. This study will serve as the primary basis for this research due to its relevance.

In parallel with DenseNet, this study also employs YOLOv11 due to its modern architecture and demonstrates effectiveness in medical imaging tasks. A 2025 research paper has shown YOLOv11 achieving high accuracy in multi-class wound classification with F1 scores of 0.836 and mAP of 0.893 [11]. YOLOv11 has also been successfully integrated into annotation-efficient pipelines for pressure ulcer segmentation and DESIGN-R scoring, requiring minimal manual labeling while maintaining robust generalization across wound types [13]. Its lightweight architecture and real-time inference speed (as low as 2.6 ms per image) further highlight its clinical practicality, particularly in resource-constrained environments [14]. By including YOLOv11 alongside DenseNet, this study compares a state-of-the-art object detection-derived model with a densely connected classification network, offering complementary perspectives on the impact of preprocessing for pressure ulcer stage classification.

Pressure ulcers, also referred to as bedsores or pressure injuries, are localized for damage to the skin and underlying tissue that occurs primarily over bony areas because of prolonged pressure, often in individuals with limited mobility [15]. Pressure ulcer severity is categorized into six stages, however, this study will only focus on the four main stages (excluding unstageable and deep tissue pressure injuries): Stage I (intact skin with non-blanchable erythema), Stage II (partial-thickness skin loss involving exposed dermis), Stage III (full-thickness skin loss), and Stage IV (full-thickness skin and tissue loss) [15] [16]. A 2020 systematic review of data from over 2.5 million patients reported that 12.8% of hospitalized adult patients are affected by pressure injuries, highlighting the global scale of the problem and the urgency for improved clinical interventions [16]. Despite being preventable, the persistence of pressure ulcers is closely tied to the quality of nursing care. Factors such as the ongoing

global shortage of nursing personnel and inadequate training in wound assessment and management among healthcare professionals continue to impede efforts in prevention and timely treatment [18] [19]. These challenges point to the need for innovative and automated solutions that can support healthcare providers in accurately identifying and managing pressure ulcer stages.

1.2 The Problem, Gap, or Opportunity

Pressure ulcer (PU) classification models have been developed using datasets of consistent, high-quality images sourced from controlled medical environments. These models often assume uniform image quality and controlled conditions that ensure high-resolution, properly lit images. However, in real-world healthcare settings, especially when images are collected across different hospitals, clinics, and online resources, the image quality varies significantly. In these settings, images may be poorly lit, blurry, or inconsistently captured, which can negatively affect the performance of traditional models. Despite their high accuracy in research settings, existing models may struggle to generalize across this variety of image qualities and data sources, leading to a potential decrease in performance when applied to real-world scenarios.

Existing studies rely on high-quality or uniformly annotated datasets, where the image quality is controlled and consistent. Most prior studies report CNN classification accuracy under these controlled conditions, but few quantify the actual impact of preprocessing techniques on model performance. This leaves an important gap in understanding how preprocessing might change outcomes when moving from expensive ideal datasets to real-world, noisy ones. Few studies also adopt a factorial design that can separate the benefit of preprocessing from the choice of model, or test whether their interaction is significant on mixed-quality data.

Moreover, while augmentation techniques like StyleGAN2-ADA have been used in other fields to generate synthetic data, their application in pressure ulcer datasets is limited [6]. Additionally, diffusion models (like DDPM) have shown promise in improving image quality by removing noise and enhancing details [5], although their application in PU image enhancement has only been explored in a small number of studies. What remains unclear is how much preprocessing, particularly through diffusion enhancement and GAN-based augmentation, improves CNN classification compared to no preprocessing, particularly when tested on mixed-quality datasets.

This study addresses these gaps with a 2×4 factorial experiment that operates across varied image qualities. We structure the evaluation as Model (DenseNet vs. YOLOv11) $\times$ Data Condition (NON-AUGMENTED, DDPM, StyleGAN2-ADA, DDPM+StyleGAN2-ADA) to test the effect of generative preprocessing rather than assume its benefit. The design quantifies the average impact of each data condition across models, ranks DenseNet and YOLOv11 under the same regimes, and tests whether any benefit from preprocessing depends on the chosen model. The goal is to generate evidence that informs pipeline design and model selection for practical, accessible wound care tools usable in real-world, mixed-quality clinical settings.

1.3 Research Questions

This study is guided by key research questions aimed at evaluating how preprocessing methods influence CNN performance in pressure ulcer stage classification. With the growing demand for accurate diagnostic tools in real-world clinical settings, the following questions were formulated:

1. How do different data settings (NON-AUGMENTED, ES-DDPM, StyleGAN2-ADA, ES-DDPM+StyleGAN2-ADA) affect classification performance of the models (DenseNet and YOLOv11)?

2. What is the limitation of the proposed augmentation models when applied to varying quality images on classifying pressure ulcer stages I to IV?

1.4 Research Objectives

This research seeks to address the challenges of classifying pressure ulcer stages from low-quality and inconsistently captured images by evaluating the effect of preprocessing on CNN performance. By comparing classification results under non-augmented and preprocessed conditions, the study aims to provide evidence of how preprocessing impacts robustness in real-world clinical scenarios.

The following objectives have been formulated to guide the study:

1. To measure how different data settings (NON-AUGMENTED, ES-DDPM, StyleGAN2-ADA, and ES-DDPM+StyleGAN2-ADA) affect classification performance in terms of accuracy, precision, recall, and confusion matrix of the models DenseNet and YOLOv11.
2. To determine the limitations of the proposed augmentation models when applied to varying quality images on classifying pressure ulcer stages I to IV.

1.5 Significance of the Study

The significance brought by the implications and contributions of this study lies in its ability to provide an assistive deep learning framework for the automated classification of pressure ulcer stages, especially from the inconsistent quality of medical images. This research explores the integration of diffusion models and generative adversarial networks into a unified architecture that aims to enhance medical images and improve classification accuracy. By doing so, it expands the application of computer vision and deep learning in healthcare, particularly in supporting clinical decision-making, improving wound assessment, and enabling more accessible

diagnostic tools. This study will be done with the aim of being of benefit to the following target beneficiaries:

Contribution to Medicine and Wound Care. With pressure ulcers posing a critical concern in healthcare, particularly for individuals with reduced mobility, this research aims to support frontline healthcare workers, especially wound care nurses, in the early detection and intervention of pressure ulcers, which is crucial for timely treatment. The conventional visual examination method for diagnosing pressure ulcer severity requires professional expertise, which can be challenging for inexperienced nurses or lay caregivers in home settings. Implementing such technologies can improve decision-making and positively impact the rate of hospital-acquired pressure injuries. Beyond improved patient outcomes, these advancements may offer practical benefits for healthcare settings by potentially reducing hospital stays and treatment costs associated with advanced pressure ulcers.

Innovation in Handling Varied Quality of Medical Images and Augmentation. Deep learning models typically require massive datasets for training, and medical data often suffers from diversity, with varying quality due to different equipment and other factors. This study aims to address the critical issue of low-quality medical images and limited datasets in deep learning applications by introducing a diffusion-based image enhancement module using DDPM to improve the quality of low-resolution and noisy pressure ulcer images, making them suitable for accurate classification. Moreover, the integration of StyleGAN2-ADA, which can generate high-quality synthetic images that resemble real medical photographs, supports robust training even with limited medical image data, overcoming the scarcity of comprehensive datasets. Together, these methods are integrated with the goal of enhancing the model's ability to generalize across real-world conditions and contribute

to the broader development of AI systems that are practical, scalable, and resilient to data variability.

Future Research or Related Studies in Computer Vision for Medical Image Classification and Assessment. Lastly, this study aims to contribute to the growing body of literature surrounding the use of computer vision in medical diagnostics, particularly in the automated assessment of skin and wound-related conditions. By testing CNNs with and without preprocessing, this research provides an evidence-based framework that future studies can extend, whether to improve preprocessing pipelines, enhance interpretability, or transfer similar approaches to other medical imaging tasks. Subsequent research may build upon the methods and findings presented in this study to improve performance, enhance interpretability, or apply similar approaches to other areas of medical image analysis, such as dermatological diseases, diabetic foot ulcers, or surgical wound monitoring. As the field of computer vision continues to evolve, the hope is that this study serves as a foundation for further innovation in assistive diagnostic technologies, addressing existing gaps and inspiring broader applications across the medical domain.

1.6 Scope and Limitations

This research focuses on the development of a deep learning framework for the automated classification of pressure ulcers, specifically targeting stages I (mild) to IV (severe), and does not include normal (no pressure injury), unstageable, or deep tissue pressure injuries. Limiting the scope to only stages I to IV provides a well-defined and visually distinguishable classification range, as unstageable, and DTPI may require manual probing or further clinical evaluation, which can introduce additional complexity.

The experimental setup covers eight conditions formed by two models and four data settings:

1. DenseNet trained on NON-AUGMENTED
2. DenseNet trained on ES-DDPM
3. DenseNet trained on StyleGAN2-ADA
4. DenseNet trained on ES-DDPM + StyleGAN2-ADA
5. YOLOv11 trained on NON-AUGMENTED
6. YOLOv11 trained on ES-DDPM
7. YOLOv11 trained on StyleGAN2-ADA
8. YOLOv11 trained on ES-DDPM + StyleGAN2-ADA

No other CNN architectures (for example, ResNet or VGG) are included. The preprocessing pipeline is limited to Early-Stopped DDPM for enhancement and StyleGAN2-ADA for augmentation, including their combined setting. Preprocessing is applied only to the training set; validation and test sets remain NON-AUGMENTED. The primary goal is to assess how these data settings influence accuracy, precision, recall, and confusion patterns under mixed-quality clinical images.

The study is limited to the classification of pressure injuries and does not include the detection or classification of other wound types, such as diabetic foot ulcers, venous ulcers, or surgical wounds. However, the methodology and architectural framework developed in this study may be adapted for use in the classification of other types of medical wounds in future research. The model is exclusively based on visual image analysis and does not incorporate additional diagnostic factors such as patient history, physiological data, or clinical risk assessment tools like the Braden Scale, which are typically used in comprehensive clinical evaluations of pressure ulcer risk and severity.

Furthermore, the study does not address the development or deployment of a real-time clinical application. Instead, it focuses on building and evaluating the model’s performance within a controlled environment. To ensure reliability and clinical relevance, the dataset used in this study will be reviewed and annotated with the guidance of a licensed professional practitioner, who will contribute to the classification of pressure ulcer stages and provide domain-specific insights essential to validating the dataset and model outcomes. This study aims to demonstrate the extent to which preprocessing pipelines affect CNN classification performance compared to non-augmented baselines, thereby clarifying whether such preprocessing is necessary in practical deployment.

To better reflect the conditions found in diverse clinical settings, the model will be trained and evaluated on images of varying quality. This includes:

- **Resolution:** Images ranging from low-resolution ($\leq 128 \times 128$ pixels) to high-resolution ($\geq 512 \times 512$ pixels) will be considered, simulating the inconsistency in imaging equipment and capture methods used in practice.
- **Noise:** Noisy images refer to those affected by random visual distortions such as speckle noise, sensor grain, or compression artifacts. These imperfections may obscure wound edges or texture and are often present in telemedicine images or older datasets.
- **Clarity:** Images with reduced clarity may exhibit blurriness due to motion, out-of-focus capture, or low camera quality. These conditions affect the model’s ability to identify wound boundaries and texture.
- **Lighting Conditions:** Lighting will vary from bright (overexposed or harsh direct light causing washed-out details) to dark (underexposed with poor visibility of wound features), as well as uneven lighting (e.g., strong

shadows or localized glare). These conditions simulate real-world clinical environments where lighting cannot always be controlled.

Targeted pressure ulcer stages included in this study according to the National Pressure Injury Advisory Panel (NPIAP) [19]:

- **Stage I:** Intact skin with non-blanchable redness, accompanied by changes in temperature, firmness, or sensation.
- **Stage II:** Partial-thickness of skin loss with exposed dermis, often appearing as a red or pink wound bed or a ruptured serum-filled blister.
- **Stage III:** Full-thickness skin loss exposing subcutaneous fat, with presence of slough, granulation tissue, and rolled wound edges.
- **Stage IV:** Full-thickness tissue loss with exposure of muscle, tendon, or bone, often involving undermining, tunneling, and necrotic tissue. [OBJ]

Chapter 2

REVIEW OF RELATED LITERATURE & STUDIES

This chapter presents a comprehensive review of existing research on the automated classification of pressure ulcer severity using deep learning techniques. It begins by establishing a medical foundation on pressure ulcers, followed by an examination of Denoising Diffusion Probabilistic Models (DDPMs), StyleGAN2-ADA, and Convolutional Neural Networks (CNNs). The chapter also highlights key related studies and explains how the selected model in this study aims to address persistent challenges in clinical image classification.

2.1 Pressure Ulcers

2.1.1 Overview of Pressure Ulcers

Pressure ulcers, otherwise known as bedsores, pressure injuries, or decubitus ulcers, are defined as areas of damaged skin and underlying tissue resulting from

prolonged pressure that restricts blood flow [11]. These injuries typically develop on bony prominences such as the tailbone, heels, elbows, and hips, especially in individuals with limited mobility who cannot easily change positions, such as bedridden patients [11]. Common pressure points in a lying position include the back of the head, shoulder blades, elbows, tailbone, and heels (see Figure 1), while in a seated position, high-risk areas shift to include the ischium, or the lower part of the hip bone, and remain concentrated on the back of the head, shoulder blades, tailbone, and heels (see Figure 2) [20].

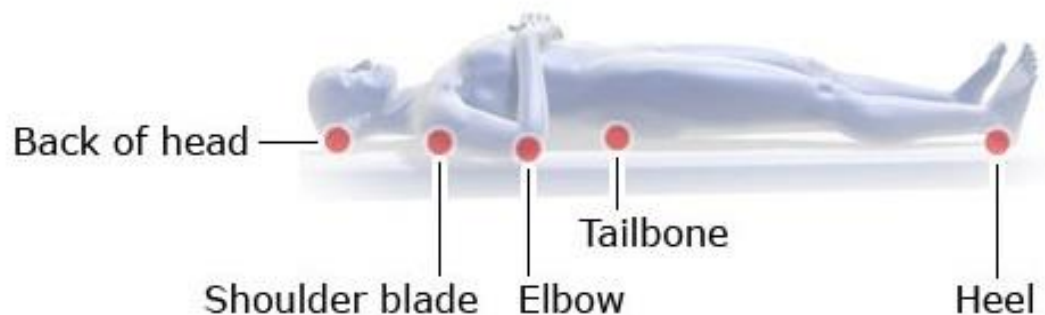


Figure 1. Common Pressure Points in a Lying Position.

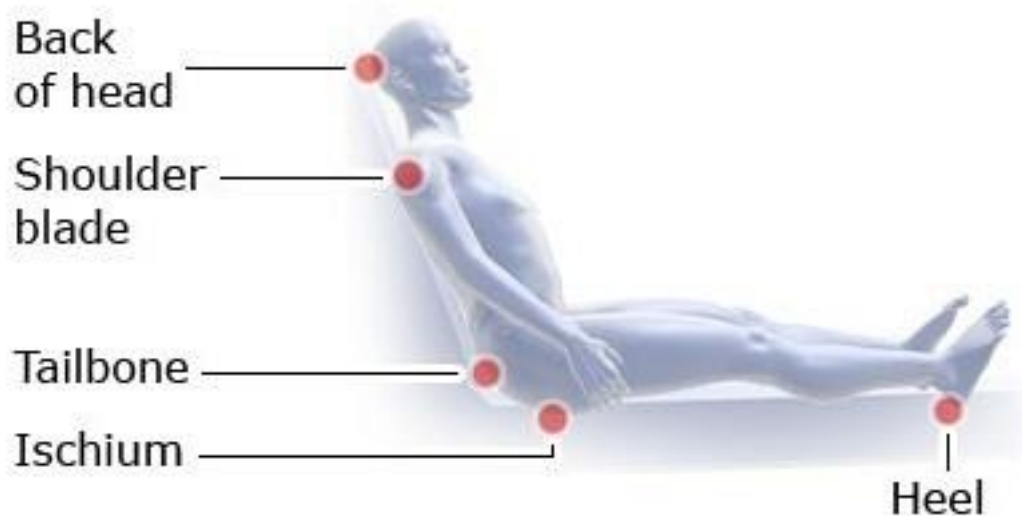


Figure 2. Common Pressure Points in a Seated Position.

Contributing factors to pressure ulcer development include impaired sensory perception (e.g., spinal cord injuries, neurological disorders), poor nutrition and hydration, conditions affecting blood flow (e.g., diabetes and vascular disease), and advanced age [21]. If left untreated, pressure ulcers can lead to severe complications such as intense pain, localized and systemic infections, cellulitis, osteomyelitis, and in extreme cases, even sepsis and death. These complications can significantly prolong hospital stays, increase healthcare costs, and drastically diminish a patient's quality of life [11]. The progression of pressure ulcers can be classified into stages that reflect the depth of tissue damage, ranging from superficial skin redness to deep wounds exposing muscle or bone [11]. Figure 3 illustrates the four primary stages of pressure ulcer development in relation to skin, soft tissue, and bone.

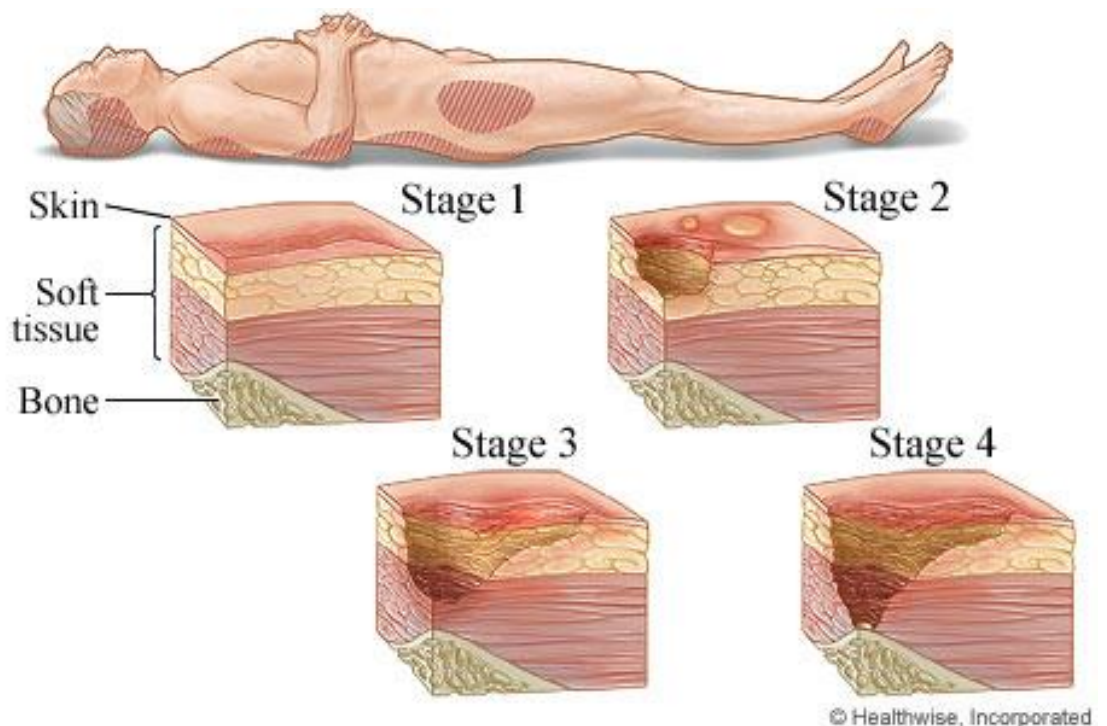


Figure 3. Visual Progression of Pressure Ulcer Stages.

The occurrence of pressure ulcers is often considered a critical indicator of the quality of nursing care, as these injuries are preventable with diligent attention and appropriate interventions [17]. Through vigilant skin assessment, timely repositioning, and proper nutrition healthcare providers can effectively prevent the development and progression of these injuries [17]. Preventing pressure ulcers is not only crucial for patient well-being but also reflects a high standard of clinical care [17].

2.1.2 Pressure Ulcer Stages

To manage and treat pressure injuries effectively, healthcare professionals rely on a standardized clinical staging system that classifies pressure ulcers based on the depth and extent of observable tissue damage [24]. This classification is essential for guiding clinical decision-making, determining the severity of the wound, and planning appropriate interventions [24]. Additionally, in the context of Convolutional Neural Networks (CNNs), this system provides the necessary categorical labels for training models to perform accurate stage classification from wound images [25].

The NPIAP staging system, developed by the National Pressure Injury Advisory Panel (NPIAP), is internationally recognized and widely adopted in clinical practice which serves as the foundation for pressure injury classification, particularly in the United States [19]. The European Pressure Ulcer Advisory Panel (EPUAP) and the Pan Pacific Pressure Injury Alliance (PPPIA) collaborate with NPIAP in publishing joint clinical practice guidelines through the International Guideline Governance Group (IGGG) [26]. This globally recognized system ensures consistency in terminology, staging, and treatment standards across healthcare systems, reinforcing the validity of using NPIAP-based classifications in deep learning applications [24]. This system includes six clearly defined categories: Stage 1 through Stage 4, Unstageable Pressure Injury, and Deep Tissue Pressure Injury (DTPI) [26]. Each stage is defined by specific

visual characteristics related to tissue involvement, making it ideal for both clinical use and image-based classification tasks in machine learning models. Visual illustrations of each stage have also been created to assist in training, teaching, and understanding the extent of tissue involvement (See Figure 4) [24].

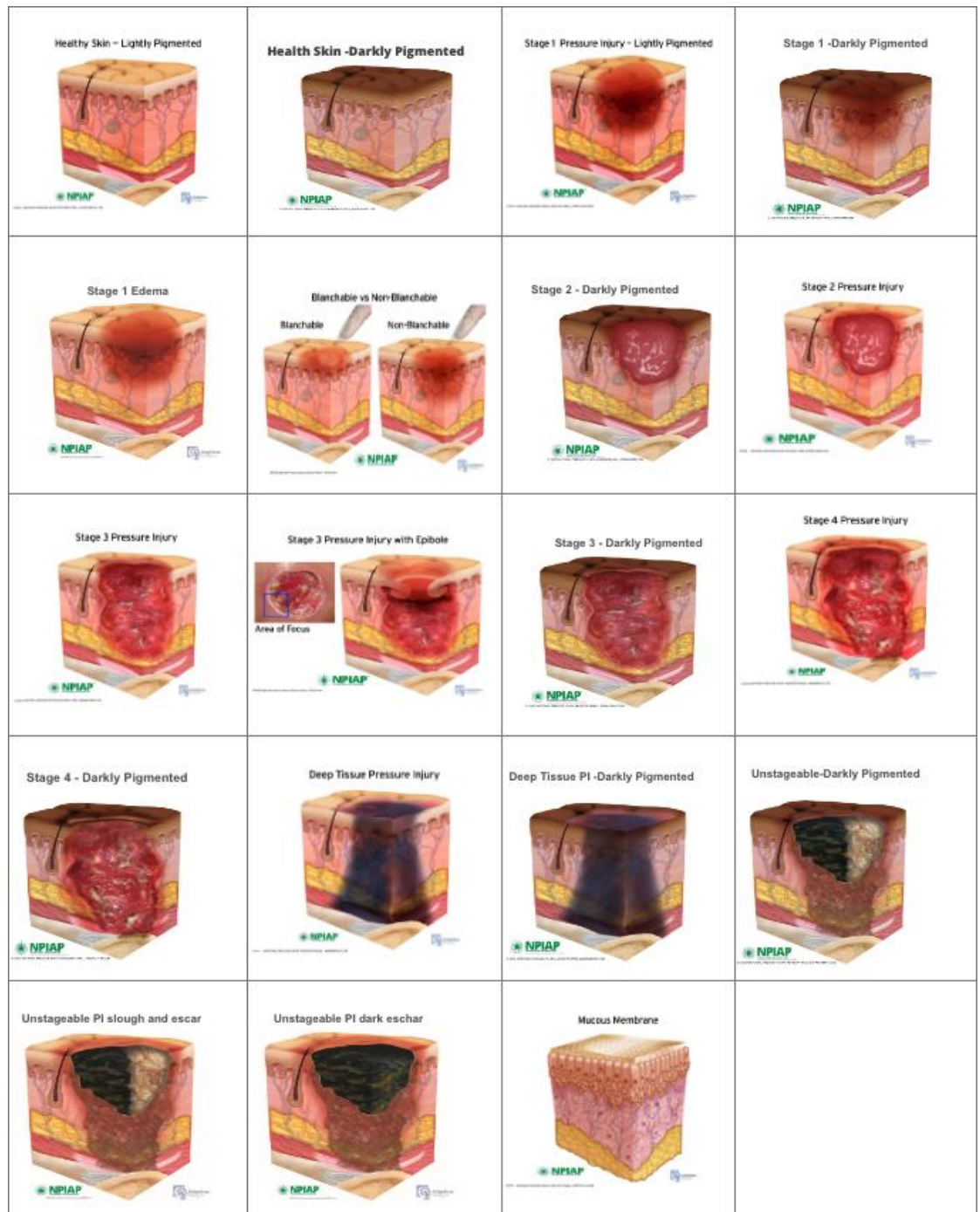


Figure 4. Visual Illustrations of Each Stage from the NPIAP.

In contrast, risk assessment scales, such as the Braden, Norton, and Waterlow scales, are designed to predict the likelihood of pressure ulcer development by evaluating various patient conditions and risk factors including sensory perception, moisture, activity level, mobility, nutrition, friction, and shear (internal tissue stress caused by the skin and deeper tissues moving in opposite directions) [28]. While clinically valuable, these tools rely on subjective assessments and clinical data inputs that cannot be extracted from images alone. Therefore, they are better suited for Electronic Health Record (EHR)-based predictive systems, not vision-based classification models like CNNs [29].

Despite the widespread adoption of the NPIAP system, accurate staging remains a clinical challenge, even for experienced practitioners [31]. Studies have shown that classification reliability varies significantly among clinicians, and disagreements have been observed between bedside nurses and wound care specialists [32]. Common sources of confusion include differentiating Stage 1 erythema from early signs of deep tissue injury, and distinguishing Stage 2 partial-thickness wounds caused by friction or moisture from pressure-induced lesions [31]. Such inconsistencies highlight the limitations of manual staging and underscore the potential of AI-based models to enhance objectivity, reproducibility, and diagnostic accuracy in pressure ulcer assessment.

The NPIAP staging system defines six distinct categories of pressure injury, each characterized by specific tissue involvement [26]. To support accurate clinical assessment, sample images illustrating each stage, shown in the sequence of older adults, Asian skin, dark skin and light skin tones, are taken from the Pan Pacific Pressure Injury Alliance (PPPIA) [33]. These categories provide a comprehensive framework for assessing the severity and extent of tissue damage [19]:

Stage 1 Pressure Injury: Non-blanchable erythema of intact skin

Stage 1 is identified by intact skin that shows a localized area of non-blanchable redness, which may appear differently depending on skin pigmentation. Before any visible skin damage occurs, changes in sensation, firmness, or temperature might be detected. Importantly, dark red or purple discoloration is not part of Stage 1, as this could indicate a deeper tissue injury instead.

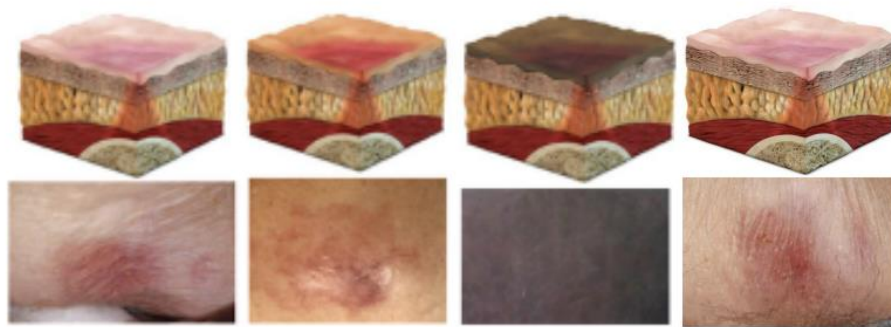


Figure 5. Sample Images of Stage I Pressure Injuries.

Stage 2 Pressure Injury: Partial-thickness skin loss with exposed dermis

Stage 2 pressure injuries involve superficial skin damage, characterized by partial loss of the epidermis and/or dermis. The wound typically appears pink or red, moist, and may present as a ruptured or intact serum-filled blister. At this stage, deeper tissues like fat are not visible, and there is no presence of granulation tissue, slough, or eschar.

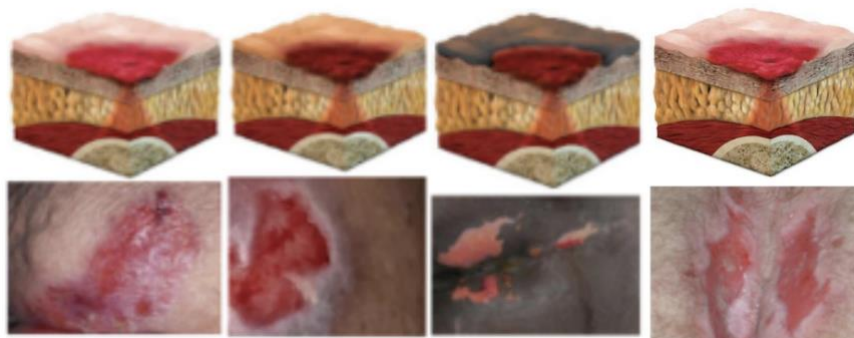


Figure 6. Sample Images of Stage II Pressure Injuries.

Stage 3 Pressure Injury: Full-thickness skin loss

In Stage 3, the ulcer extends through the full thickness of the skin, often exposing subcutaneous fat. Granulation tissue and rolled wound edges (epibole) are commonly seen, and there may be visible slough or eschar. The depth of the wound varies by location on the body and can involve undermining or tunneling, but deeper structures like muscle, tendon, or bone remain covered.

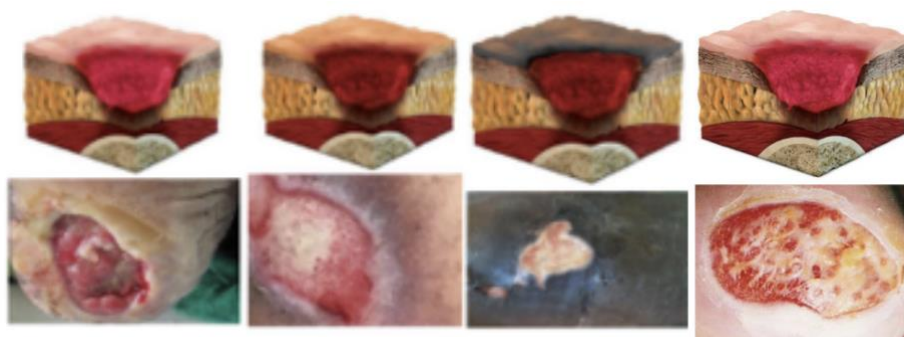


Figure 7. Sample Images of Stage III Pressure Injuries.

Stage 4 Pressure Injury: Full-thickness skin and tissue loss

Stage 4 injuries show extensive tissue damage, with full-thickness skin and tissue loss that exposes or allows direct palpation of underlying structures such as muscle, tendon, ligament, cartilage, or bone. Slough and eschar may be present, and features like epibole, undermining, and tunneling are typically observed. The depth of the injury varies depending on the affected anatomical site.

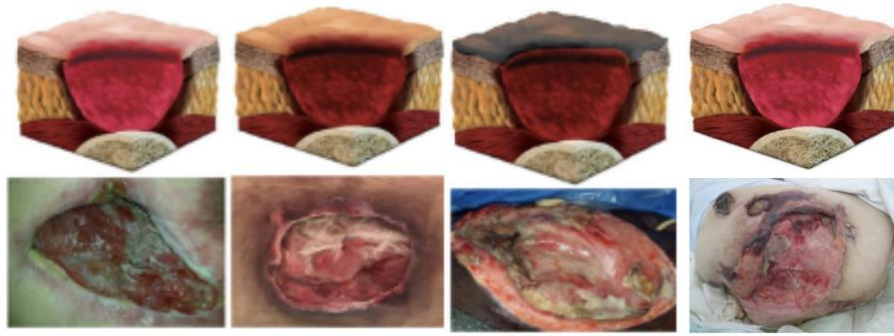


Figure 8. Sample Images of Stage IV Pressure Injuries.

Unstageable Pressure Injury: Obscured full-thickness skin and tissue loss

Unstageable pressure injuries involve full-thickness tissue loss; however, the true extent of the damage cannot be assessed because the wound is covered by slough or eschar. Once this material is removed, the injury is often found to be at Stage 3 or 4. If the eschar is stable—dry, adherent, and intact—particularly on ischemic limbs or heels, it should not be disturbed.

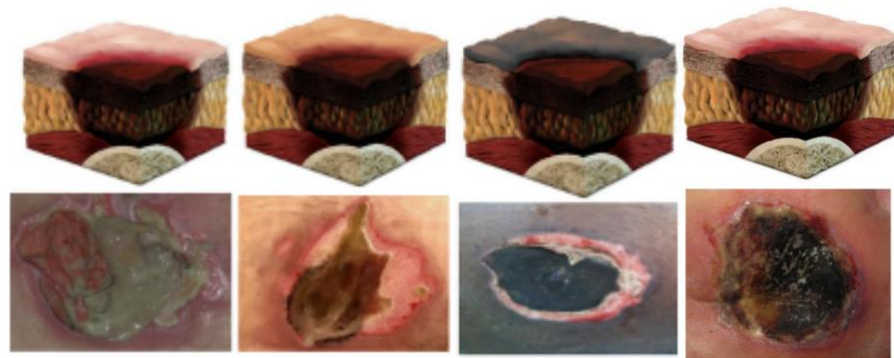


Figure 9. Sample Images of Unstageable Pressure Injuries.

Deep Tissue Pressure Injury: Persistent non-blanchable deep red, maroon, or purple discoloration

A deep tissue pressure injury presents a localized area of persistent, non-blanchable dark red, maroon, or purple discoloration of the skin. The skin may be intact

or already broken, sometimes revealing a dark wound bed or a blood-filled blister. This type of injury results from prolonged pressure and shear at the bone-muscle interface and may rapidly progress or resolve without surface tissue loss.

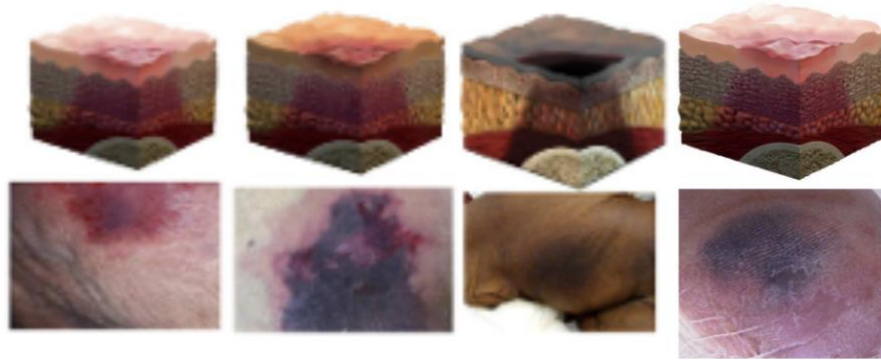


Figure 10. Sample Images of Deep Tissue Pressure Injuries.

Accurate staging is not only critical for effective clinical decision-making and patient care but also plays a foundational role in the development of automated classification systems [34]. In machine learning models, the accuracy and consistency of ground truth labels directly influence model performance [34]. This consistency enables AI models to learn distinct visual features associated with each stage, improving their ability to generalize across patient populations and healthcare settings [37]. As such, the NPIAP system serves a dual purpose, guiding clinical interventions and enabling the creation of reliable and reproducible pressure injury classification tools.

2.1.3 Prevention, Management, and Global Burden of Pressure Ulcers

Addressing pressure ulcer care and treatment remains a complex and resource-intensive challenge in modern healthcare. Prevention is the most effective form of treatment, requiring continuous and coordinated care, often through structured programs that address individual risk factors and promote best practices [38]. Core

strategies include routine repositioning (typically every two hours), the use of pressure-relieving surfaces, and nutritional support to maintain skin integrity [38]. However, once a pressure ulcer develops, care becomes more intensive, requiring wound management, infection control, and nutritional interventions to support healing [39]. These demands highlight the critical role of nurses who are on the frontline of both prevention and treatment [17]. Yet, effective pressure ulcer care requires a multidisciplinary approach involving collaboration among nurses, physicians, physical therapists, and dietitians to apply clinical guidelines effectively [41]. The difficulty in maintaining such consistent, round-the-clock care contributes to the widespread nature of the problem, particularly in healthcare systems struggling with staffing shortages [17]. Pressure ulcers persist as a significant global health issue with a worldwide prevalence of estimated 12.8% among hospitalized adult patients according to a large-scale review involving over 2.5 million cases, and even higher rates in long-term care facilities and intensive care units where patients often have limited mobility and complex medical needs [16]. In the United States, this challenge is particularly evident as incidence rates have remained stable for three decades with rates of approximately 10.38 per 100,000 population in 1990, dropping only slightly to 8.63 in 2017, with no significant change observed across age or sex (See Figure 11 and Figure 12) [42].

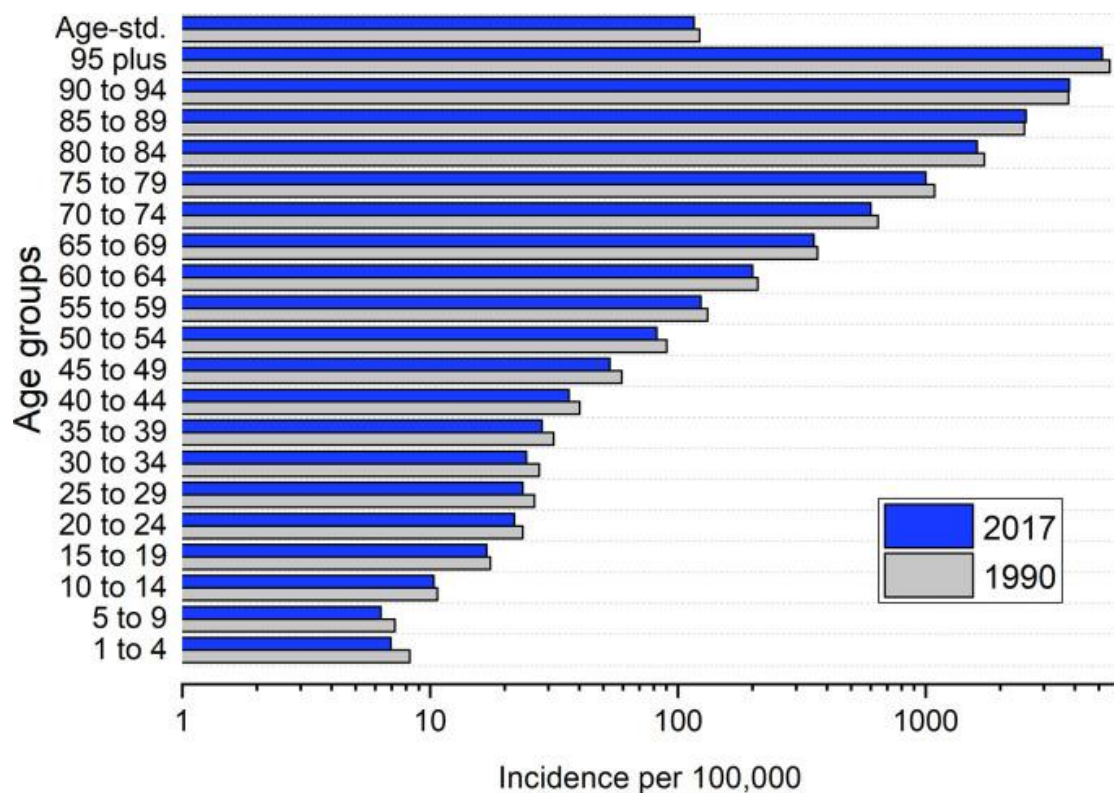


Figure 11. Incidence per 100,000 populations in various age groups.

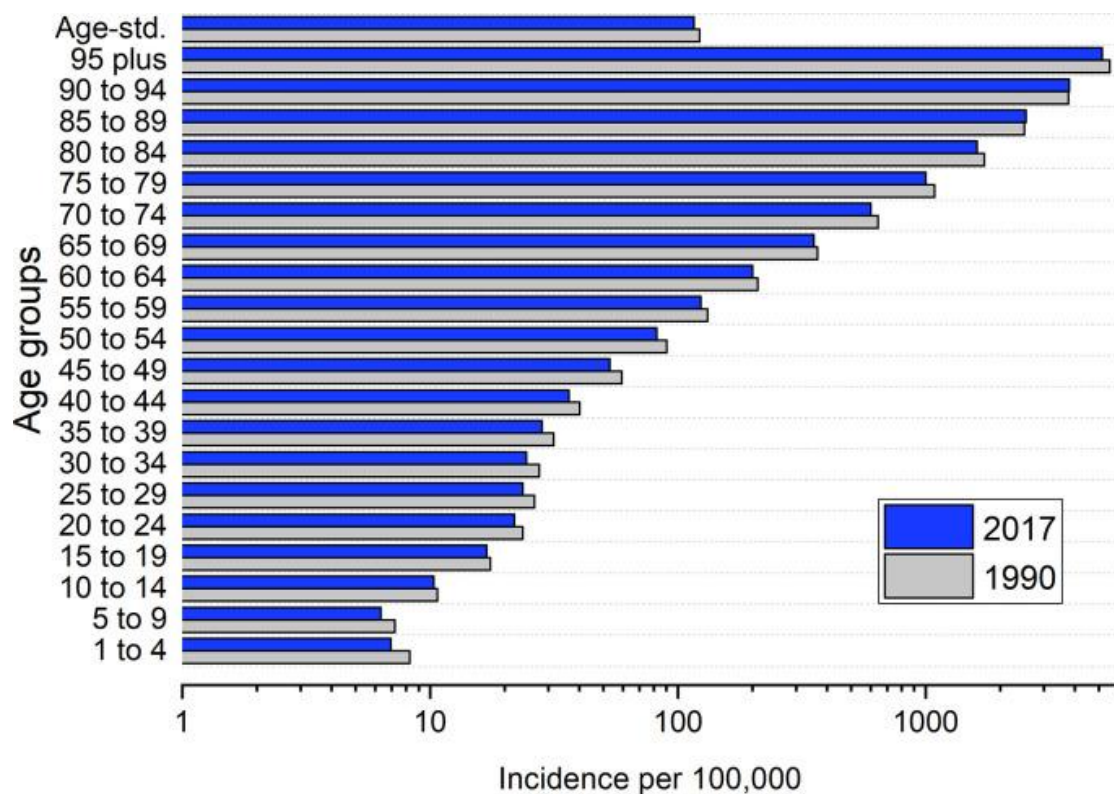


Figure 12. Incidence per 100,000 population in male and female patients.

In the Philippines, the incidence of pressure injuries among orthopedic patients aged 19 years and older was reported to be 20%, with many patients who developed pressure injuries were 24-33 years old [43]. A study conducted from January 2020 to December 2021 at Quirino Memorial Medical Center in Quezon City, Philippines, found an incidence rate of hospital-acquired pressure injuries (HAPI) of 1.4% among older patients, equivalent to 14 new cases per 1,000 population [43]. Within hospitalized orthopedic patients in the Philippines, impaired sensory perception and constant moisture were identified as significant risk factors for pressure ulcer development [44]. This widespread prevalence is further exacerbated by a persistent global shortage of nurses, as acknowledged by the World Health Organization, which hinders the ability of healthcare systems to deliver timely interventions and sustain preventive measures, leading to avoidable wounds becoming chronic and resource-intensive complications [17].

2.2 Denoising Diffusion Probabilistic Model (DDPM)

2.2.1 Overview of Diffusion Models and DDPM

Diffusion models are a generative technology that has gained significant popularity in recent years due to their flexible high-dimensional data modeling and guided sample generation tailored to specific tasks [46]. Originally inspired by concepts from non-equilibrium thermodynamics, these models work by gradually adding Gaussian noise to training data in a “forward diffusion process” until the data becomes pure noise [47]. The central idea is to then reverse this noising process, progressively transforming random noise back into a coherent and realistic data sample [47]. Diffusion models have achieved impressive results across various domains, including computer vision, audio, reinforcement learning, computational biology, speech synthesis, and natural language processing [46].

As probabilistic models, diffusion models simulate a stochastic diffusion process to generate realistic outputs [49]. This framework involves two key stages: a forward process, where data is incrementally corrupted with noise, and a reverse process, where the model learns to remove this noise to recover the original data distribution.

Forward Diffusion Process

According to O'Connor [47], during the forward process, Gaussian noise is progressively added to input data point (e.g., an image x_0), corrupting it over a series of T timesteps until it approximates pure isotropic Gaussian noise. This is modeled as a Markov chain with a defined variance schedule $(\beta_1, \dots, \beta_T)$, which controls the amount of noise added at each timestep, determining how fast or slow the input data is corrupted. After this transformation, the original data becomes indistinguishable and is transformed into pure Gaussian noise. This process is illustrated in Figure 13 and Figure 14.

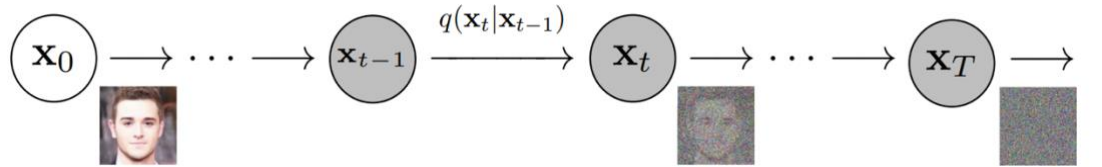


Figure 13. Diagram of the Forward Diffusion Process.



Figure 14. Sample of the Forward Diffusion Process.

Reverse Diffusion Process

As explained by O'Connor [47], the objective of a diffusion model is to learn the reverse process, where the machine learning takes place as the model learns to iteratively remove noise and reconstruct the original data, transforming pure Gaussian noise back into a clean image. Like the forward process, the denoising process is also modeled as a Markov chain, where the model estimates the probability distribution $p_\theta(x_{t-1}|x_t)$ at each timestep. Instead of reconstructing the original image directly, the model is trained to predict the noise applied incrementally across each step. This learned noise estimation enables the model to recover the underlying data structure from corrupted inputs. The entire denoising process transforms a simple distribution (Gaussian noise) into a complex one (natural-looking image). An example of this reverse process is shown in Figure 15.

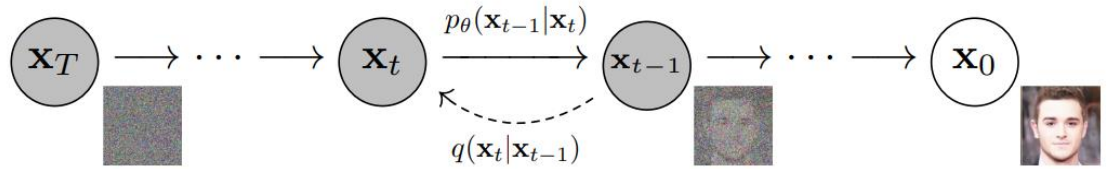


Figure 15. Diagram of the Reverse Diffusion Process.

Denoising Diffusion Probabilistic Models (DDPMs), originally introduced in 2015 and popularized by Ho et al. in 2020, are a specific implementation of diffusion models and are categorized as latent variable models, meaning that they are a type of machine learning models that involve hidden or “latent” variables that are not directly observed in the data but are inferred [5]. What distinguishes DDPMs from other diffusion-based models is their use of a fixed variance schedule and a training objective centered on noise prediction [51]. Instead of learning to reconstruct the original image directly, the model is trained to predict the exact noise that was added at each timestep

[50]. This approach exemplifies how diffusion models can achieve high-quality image generation without relying on adversarial training, offering a more stable and scalable alternative to Generative Adversarial Networks (GANs) [52]. In this study, DDPM is selected for its capacity to denoise and reconstruct clearer, high-resolution versions of low-quality pressure ulcer images. This enhancement step is expected to support improved performance in subsequent classification tasks, as will be discussed in Chapter 3.

2.2.2 Applications of DDPMs in Image Processing and Healthcare

DDPMs have quickly become a cornerstone in modern generative modeling, demonstrating significant capabilities across a spectrum of image processing tasks [53]. By learning to reverse a gradual noise corruption process, DDPMs can synthesize high-fidelity images from pure Gaussian noise, making them particularly useful in image restoration and generation [54]. Beyond their success in general image processing domains such as super-resolution and inpainting, DDPMs have also found valuable applications in medical imaging, where they address key challenges like limited data availability, noise suppression, and low-resolution enhancement [55]. Their growing impact across both synthetic and clinical image domains highlights their relevance for practical, real-world scenarios, especially in contexts where image quality directly influences decision-making and downstream analysis [56].

Demonstrating remarkable performance across fundamental image processing tasks, DDPMs have been successfully applied to image generation, super-resolution, inpainting, and noise removal, surpassing some traditional techniques and may even outperform advanced models such as Generative Adversarial Networks (GANs) [57]. Recent studies highlight DDPM's capacity for high-resolution image synthesis with complex features, making them strong contenders in the field of deep learning and

computer vision [58]. The ability of DDPMs to generate diverse and high-fidelity images stems from their inherent stability and consistency in handling complex data distributions [59].

In image generation, DDPMs have revolutionized the creation of synthetic images that are indistinguishable for real ones [53]. This has been particularly beneficial in domains where collecting large, diverse datasets is challenging, such as certain medical fields [57]. For example, a 2024 study by Kim et al. explored the feasibility of using DDPMs to synthesize retinal fundus photographs from a small dataset and found that the model successfully generated synthetic images despite limited data, highlighting the potential of DDPMs for domain-specific image synthesis [58]. Beyond unconditional generation, DDPMs have also been used to generate images conditioned on specific inputs, allowing for a more controlled and targeted synthesis [60].

Image super-resolution is a common image enhancement technique aimed at enhancing the resolution of an image while preserving its original content and fine details, resulting in a clearer, high-resolution version of the input [61]. DDPMs have been successfully applied to this task and have shown promising results by overcoming common issues in generative models such as over-smoothing, where output images appear overly smooth or blurry, and mode collapse, where the model fails to produce diverse outputs [61]. Recent advancements include an Adaptive Semantic-Enhanced DDPM (ASDDPM), which improves high-frequency detail alignment by incorporating low-frequency semantic features, and Fast-DDPM, which accelerates training and sampling while maintaining high generation quality for multi-image super-resolution tasks [61]. Additionally, a 2023 study by Wu et al. proposed a new DDPM-based framework for brain MRI super-resolution integrating self-attention mechanisms [63]. This makes the model more efficient by focusing on important details and reducing

unnecessary computations, further demonstrating the model’s adaptability in medical imaging contexts.

DDPMs have also been shown to excel at image inpainting, the task of filling in missing or corrupted parts of an image by reconstructing missing regions with contextual coherence and realism [64]. For instance, a 2022 study by Rouzrokh et al. demonstrated the use of DDPMs for multitask inpainting on brain MRI images, showing that they can generate diverse and high-quality reconstructions across various imaging sequences, making them highly effective for inpainting compared to GANs which often suffer from limited output diversity [65].

DDPMs have also exhibited robust performance in image denoising tasks, where the goal is to restore clear, high-quality images from noisy inputs while preserving essential visual details [62]. This capability is especially valuable in fields like medical imaging, where modalities such as ultrasound images are frequently degraded by acoustic and electronic noise [66]. A 2023 study on deep ultrasound denoising using diffusion probabilistic models showed that DDPMs can iteratively enhance image quality by effectively removing noise while retaining informative textures critical for diagnostic interpretation [66]. Notably, these models can be trained in an unsupervised manner, eliminating the need for annotated training data [66]. Quantitative results from the same study demonstrated that DDPMs outperform conventional denoising approaches in metrics such as Peak Signal-to-Noise Ratio (PSNR) and Generalized Contrast-to-Noise Ratio (GCNR). These findings highlight the practicality and robustness of DDPMs in real-world medical imaging scenarios, where preserving diagnostic detail while minimizing noise is essential.

Medical imaging plays a critical role in modern healthcare, serving as a foundation for accurate diagnosis, clinical decision-making, and patient monitoring

[67]. The enhanced image quality and dataset augmentation capabilities offered by DDPMs have the potential to significantly improve the performance of diagnostic models [56]. By generating realistic and diverse synthetic images, DDPMs can help address familiar challenges such as limited data availability and class imbalance, particularly relevant in conditions like pressure ulcers where annotated data is scarce [56]. As noted by Mahaulpatha [67], the ability of DDPMs to produce high-fidelity medical images can indirectly support the development of more accurate classification algorithms by providing richer, more representative training data [67]. This not only enhances the reliability of machine learning models but also contributes to more informed clinical decisions and better patient outcomes.

However, while DDPMs offer significant advantages in terms of output quality and stability, they are not without limitations. A primary drawback is their slow sampling speed, which results from the iterative nature of the reverse diffusion process [64]. Though recent advancements such as Fast-DDPM and other acceleration techniques aim to address this issue [62]. Despite this, the quality and flexibility of DDPM-generated data make them highly valuable in applications where fidelity is critical.

In this study, DDPMs will be applied to streamline pressure ulcer images of varying quality, which often result from differences in lighting, device resolution, and clinical setting. By enhancing these images, the goal is to produce a more consistent dataset that supports more reliable stage classification across stages I to IV. This preprocessing step is intended to reduce variability and improve the performance of the CNN-based classifier, especially when working with inconsistent real-world images.

2.2.3 Early Stopped DDPM (ES-DDPM)

The following description of Early-Stopped Denoising Diffusion Probabilistic Model (ES-DDPM) is based on the work of Lyu et al. [68], who proposed it as a strategy to accelerate the training and sampling process of DDPMs. Early Stopped Denoising Diffusion Probabilistic Model (ES-DDPM) is a strategy proposed to accelerate the training and sampling process of DDPMs. While DDPMs achieve high performances in various generation tasks, they often require hundreds or thousands of denoising steps to produce high-quality samples from Gaussian noise that may lead to significant inference inefficiency. ES-DDPM tackles this issue by halting the diffusion process prematurely after the first few steps, and then initiating the reverse denoising from a non-Gaussian distribution. This differs from a traditional DDPM that typically starts the reverse denoising process from pure Gaussian noise, making it computationally expensive due to the substantial number of denoising steps required. The method of ES-DDPM is particularly suitable for applications where enhanced clarity and uniformity of generated images are prioritized over photorealism, as it can significantly reduce computational costs while improving sample quality.

As outlined by Lyu et al. [68], ES-DDPM modifies the standard denoising diffusion process by halting the forward diffusion at an intermediate timestep $t_s < T$, where T is the total number of diffusion steps in a full-length DDPM. It begins the reverse sampling from an intermediate noised version of the image denoted as x_{t_s} , instead of starting from a completely random noise x_T like in standard DDPM. Here, t_s is a smaller timestep (e.g., 200 instead of 1000), meaning the image has only been partially noised. This intermediate input can be generated from another model such as VAEs (Variational Autoencoders) or GANs (Generative Adversarial Networks), and

then the DDPM performs the remaining denoising steps from t_s down to 0. This process is illustrated in Figure 16 [7].

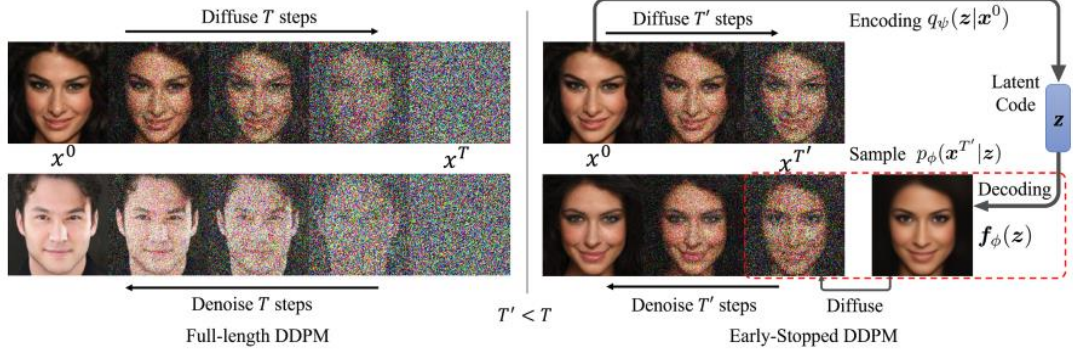


Figure 16. Comparison of full-length DDPM and ES-DDPM.

Several diffusion model variants and acceleration techniques have been developed with goals like those of Early-Stopped DDPM, aiming to improve diffusion model efficiency. For instance, Fast-DDPM is an efficient alternative for DDPMs that may reduce sampling time by 99% and training time by 80% compared to traditional DDPMs by using only 5–10-time steps instead of 1,000 steps [69]. Similarly, Partial Diffusion Models (PDMs) have been introduced for tasks like MRI super-resolution, diffuse the image using only a portion of the denoising steps, significantly reducing computational cost while maintaining competitive super-resolution quality [70]. Additionally, Denoising Diffusion Implicit Models (DDIMs) provide a more efficient class of iterative implicit probabilistic models that use the same training procedure as DDPMs but can produce high-quality samples 10 to 50 times faster using timestep-respacing strategies that allow sampling from a reduced set of timesteps using larger intervals [71]. While all these offers impressive performance, this study adopts ES-DDPM for its ability to accelerate both training and inference while maintaining compatibility with pre-trained generators such as StyleGAN2 [7].

In medical imaging applications where image clarity, noise suppression, and computational efficiency are important, ES-DDPM offers several practical advantages

[72][73][72]. It enhances visual quality by removing irrelevant noise and smoothing lighting or color inconsistencies, helping to emphasize high-level structural patterns such as wound boundaries or lesion textures [74][75]. These features are critical in diagnostic tasks like pressure ulcer classification, where subtle differences in image content can indicate clinically distinct stages [76]. The reduced number of sampling steps lowers computational demands and accelerates preprocessing, making ES-DDPM suitable for deployment in clinical or embedded environments with limited resources. For tasks where photorealism is not essential, such as classification or segmentation, early stopped generation can provide a favorable balance between clarity and efficiency. In this study, ES-DDPM is used as a preprocessing step to enhance pressure ulcer images of varying quality, aiming to produce more uniform and denoised inputs for the CNN classifier discussed in Chapter 3.

2.3 StyleGAN2-ADA (Adaptive Discriminator Augmentation)

2.3.1 Overview of GANs and StyleGAN2

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. in 2014, are a powerful class of deep generative models designed to learn patterns in data and generate new examples that closely resemble the original dataset [77][78]. The GAN framework consists of two neural networks: a generator, which creates synthetic data from random noise, and a discriminator, which evaluates whether the input data is real or generated [79]. These networks are trained in a competitive process where the generator attempts to fool the discriminator, while the discriminator learns to better detect fake inputs [80]. Over time, the generator improves its ability to produce highly realistic data, reaching a point where the discriminator can no longer reliably distinguish between real and fake inputs [79]. Ultimately reaching a point where the discriminator can no longer reliably distinguish between real and fake inputs, indicating

a well-trained generator [79]. GANs have been widely used in image and video synthesis, music composition, text generation, and data augmentation [81].

Despite their strengths, early GAN models suffer from several limitations, including mode of collapse, training instability, and vanishing gradients [80]. Mode collapse occurs when the generator produces a narrow range of outputs, failing to represent the full diversity of the dataset [80]. Instability can arise when the discriminator becomes too strong, preventing the generator from learning effectively [80][81]. Vanishing gradients may also hinder training, especially in deeper networks, by reducing the strength of weight updates in early layers [80].

To address these issues, NVIDIA introduced StyleGAN in 2018 and StyleGAN2 in 2020, which are advanced GAN architectures designed for higher-quality and more controllable image synthesis [82][83]. StyleGAN enhances image quality by incorporating principles from style transfer, enabling finer control over image features through a disentangled latent space [80][84]. This allows for manipulation of both high-level attributes (e.g., pose, identity) and low-level details (e.g., freckles, hair) [80]. However, StyleGAN commonly produces visual artifacts, particularly bob-like anomalies, that appear in the generated images which can be seen in Figure 17 [85].



Figure 17. Visual Anomalies in StyleGAN-Generated Images.

StyleGAN2 further improved image fidelity by reformulating the generator architecture, eliminating artifacts, and replacing Adaptive Instance Normalization

(AdaIN) with weight demodulation [86]. This change led to more stable training and better feature alignment. StyleGAN2 also introduced path length regularization, improving consistency and texture sharpness in the generated images [87]. This concept is demonstrated in Figure 18, where images combine coarse styles from Source B (top row) with finer styles from Source A (left column), allowing fine-grained control over features like identity and texture [85].



Figure 18. Style Mixing in StyleGAN Using Coarse and Fine Latent Codes.

Thanks to their improvements in image quality and control, StyleGAN and StyleGAN2 have been widely adopted across diverse fields, from generating photorealistic human faces and digital artwork to expanding medical imaging datasets [88]. In healthcare, these models have been used to synthesize clinically relevant medical images, helping to address data scarcity while preserving patient privacy [88]. In creative fields, they enable character design in video games, concept art generation, and even 3D modeling [89]. These broad applications underscore the versatility and

growing significance of StyleGAN-based models, particularly in settings that demand both visual realism and data diversity.

2.3.2 Adaptive Discriminator Augmentation (ADA)

Data augmentation plays a critical role in GAN training, especially when working with limited datasets. GANs require large and diverse datasets to generate high-resolution outputs and collecting a huge quantity of data for training models is often time-consuming, resource-intensive, and impractical in many real-world domains [18]. This challenge is evident in real-world applications like medical imaging, where manually annotated data is scarce and expensive to obtain [90]. When trained with insufficient data or imbalanced datasets, GANs typically suffer from discriminator overfitting, a situation where the discriminator memorizes the training images instead of learning generalizable patterns, leading to mediocre performance on unseen data [91]. This overfitting impedes the generator’s learning process and can lead to mode collapse, where the generator produces only a limited variation of outputs [18].

Adaptive Discriminator Augmentation (ADA) addresses this issue by stabilizing GAN training in data-limited scenarios. Introduced by Karras et al. from NVIDIA as part of the StyleGAN2-ADA framework, ADA enables successful training using only a few thousand images, achieving results comparable to standard StyleGAN2 models trained on much larger datasets [91] [92]. This adaptive strategy broadens the potential use of GANs in fields where acquiring large datasets is difficult.

The following discussion on ADA is primarily based on the work of Beres [93], which outlines its adaptive mechanisms, training advantages, and implementation details. Unlike traditional approaches that apply fixed augmentations, ADA works by dynamically adjusting the probability of augmentation based on the discriminator’s performance. It monitors the discriminator’s accuracy on real images and uses a control

mechanism to maintain this accuracy within a target range (typically 80-90%). This adaptive tuning ensures that the discriminator continues learning generalizable features rather than memorizing training data. Importantly, ADA prevents “leaky augmentations,” where the generator learns to mimic augmented features instead of the true data distribution, by applying augmentations only to the discriminator [94]. Another advantage mentioned is its simplicity, as it requires no changes to the network architecture or loss functions, which measure how closely a model’s predictions match the actual data, and are central to guiding the learning process [95]. ADA works seamlessly within these existing setups, making it versatile for both training new models and fine-tuning existing ones. In practice, commonly effective augmentations in data-scarce environments include *blitting* techniques (e.g., horizontal flipping and integer translation) and geometric transformations (e.g., rotation and both uniform and anisotropic scaling), which modify the image without altering its core structure. The ADA technique used in this study’s architecture is based on the investigation by Talib & Abed [96] on detecting *DeepFake* images using StyleGAN2-ADA, illustrated in Figure 19.

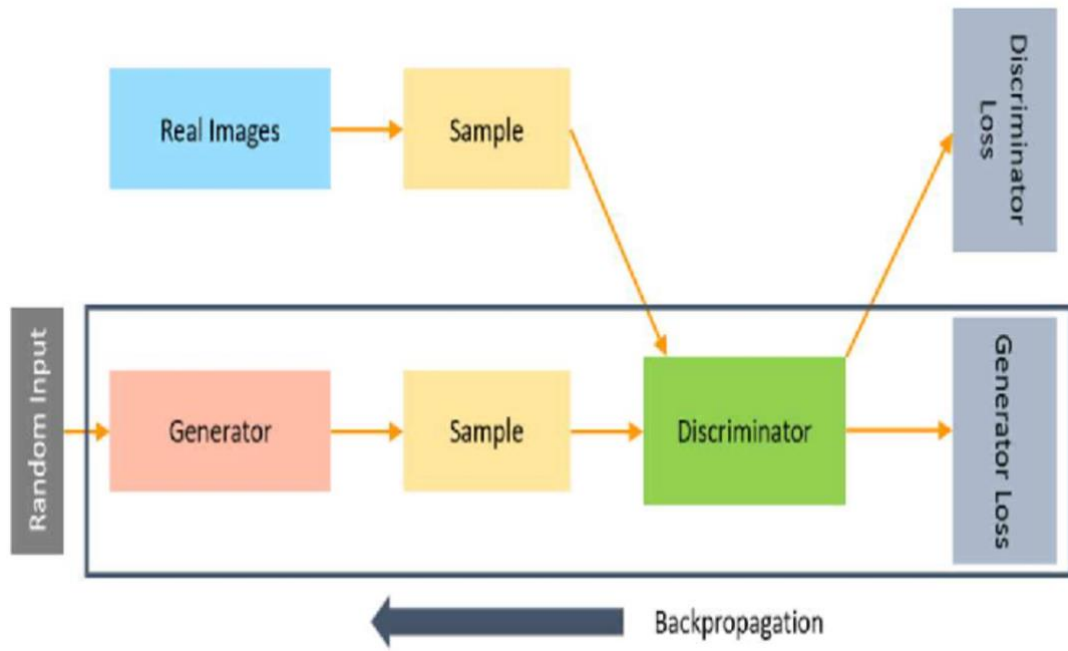


Figure 19. Adaptive Discriminator Augmentation Diagram.

2.3.3 Applications in Medical Imaging and Dataset Expansion

GANs, including StyleGAN and StyleGAN2-ADA, have proven valuable in healthcare, where medical imaging often faces challenges in obtaining large, diverse, and well-annotated datasets due to ethical restrictions like patient privacy concerns and prohibitive costs of expert labeling [97][98]. By generating realistic synthetic images, these models help overcome data scarcity and support advancements in image analysis, classification, and diagnosis [98].

A 2023 study by Skandarani et al., titled "GANs for Medical Image Synthesis: An Empirical Study," explored how different GAN architectures perform in generating synthetic medical images across multiple imaging types, including cardiac MRI, liver CT, and retinal fundus images [97]. Their findings revealed that advanced models like StyleGAN and SPADE GAN could produce highly realistic images that sometimes even fool medical experts. These results demonstrate the growing potential of GANs to address data scarcity in healthcare by generating visually convincing alternatives to real clinical data. However, the study also emphasized key limitations. Despite their visual

quality, synthetic images still fell short in clinical tasks like segmentation, where models trained on GAN-generated data performed worse than those trained on real datasets, highlighting the ongoing need for refinement in how GANs are evaluated and applied in medical contexts.

Another study, “Synthetic Data in Radiological Imaging: Current State and Future Outlook” by Sizikova et al. [99], provides a comprehensive review of how synthetic data, particularly through GANs, is transforming radiology by reducing patient risk, lowering data acquisition costs, and helping to generate large-scale and diverse datasets, especially beneficial in training machine learning models for underrepresented conditions or populations. The study outlines a wide range of use cases, including *in silico* clinical trials, algorithm development, bias mitigation, and privacy-preserving data sharing. Notably, GAN-based methods such as StyleGAN and its variants are central to this progress due to their ability to generate photorealistic medical images and annotations. However, the authors mention that significant hurdles remain; synthetic images must undergo rigorous validation to ensure realism and clinical utility, and generative models must avoid replicating training data to protect patient identity. Overall, the study underscores both the potential and the limitations of synthetic image generation.

A study by Stebbins et al. [92] titled “Enhancing Cancer Detection with StyleGAN2-ADA” demonstrated how synthetic image generation can effectively improve classification performance in cancer detection tasks with limited training data. In their experiment, StyleGAN2-ADA was applied to a small chest CT scan dataset to generate high-quality synthetic images, which were then used to augment the original data. By training a convolutional neural network (CNN) with increasing proportions of synthetic data, the researchers found that even a 25% addition of synthetic images

significantly boosted classification accuracy between normal and tumor cases. This underscores the value of StyleGAN2-ADA in expanding medical datasets, improving model generalization, and enabling deep learning in low-data settings. While the authors acknowledged that training instability and GAN quality constraints posed challenges, their findings affirm that GAN-based augmentation offers a practical path forward for enhancing medical imaging tasks

In a similar study by Fedoruk et al. [100] “Additional Look into GAN-Based Augmentation for Deep Learning COVID-19 Image Classification,” explored the effectiveness of StyleGAN2-ADA in augmenting chest X-ray datasets for COVID-19 classification tasks. Their study compared GAN-generated augmentation to classical augmentation and no augmentation across datasets of varying sizes. Results showed that GAN-based augmentation performed comparably to classical techniques in medium-sized datasets (e.g., 1000 images per class) but struggled in smaller datasets (e.g., 500 images per class), primarily due to limitations in training stability and computational cost. Despite these constraints, the study emphasized the potential of synthetic data to serve as a viable substitute for real patient data in large-scale training, especially in privacy-sensitive scenarios.

In this study, StyleGAN2-ADA will be used for data augmentation to expand the training dataset of pressure ulcer images. By generating synthetic but clinically relevant examples, the goal is to enhance classification accuracy across different pressure ulcer stages while minimizing the risk of overfitting due to limited real-world data.

2.4 Convolutional Neural Networks (CNNs)

2.4.1 Overview of CNNs

Convolutional Neural Networks (CNNs) are a type of deep learning algorithm primarily designed for object recognition tasks, including image classification, detection, and segmentation [101]. CNNs are used in various practical scenarios such as visual recognition, speech recognition, and natural language processing [102]. The architecture of CNN is inspired by the human brain's visual cortex that plays a crucial role in processing visual stimuli [102]. Simple features are extracted in early layers of this hierarchical structure, and these are used to build more complex, increasingly sophisticated representations in the deeper layers [102]. Like neurons in the visual cortex that respond to localized regions, neurons in a CNN are connected only to local receptive fields via the convolution operation, enabling more efficient feature extraction [103]. Additionally, the visual cortex neurons can detect features regardless of their location, a property mimicked by pooling layers in CNNs, which provide translation invariance by summarizing local features [104].

According to a paper by O'Shea & Nash [103], CNNs are composed of three types of layers; convolutional layers, pooling layers, and fully connected layers. Stacking these layers form a CNN architecture and its basic functionality can be broken down into four key areas:

- 1. Input Layer:** Receives the non-augmented image data represented as a multi-dimensional pixel array. It serves as the entry point of the network, preserving the spatial and depth information of the input image.
- 2. Convolutional Layer:** Applies a set of learnable filters to perform convolution operations by computing the scalar product between the filter weights and the corresponding input region which determines the output of

neurons. The resulting feature maps capture localized spatial features such as edges, textures, or patterns. After convolution, a non-linear activation function (typically the Rectified Linear Unit or ReLU) aims to apply an ‘elementwise’ activation function such as sigmoid to the output of the previous layer.

3. **Pooling Layer:** The pooling layer reduces the spatial dimensions of feature maps by *downsampling*, which lowers the number of parameters, prevents overfitting, and improves translation invariance. It works by extracting the most noteworthy features from the convoluted matrix, simplifying the representation while retaining essential information.
4. **Fully Connected Layer:** Takes the high-level feature representations from the final convolutional or pooling layers and flattens them into a one-dimensional vector. This vector is then passed through dense layers to compute class scores, enabling the final classification decision. Each neuron in a fully connected layer is connected to all activations in the previous layer, allowing for global integration of spatial information.

A basic CNN architecture is visually represented in Figure 20. Schematic Diagram of Basic CNN Architecture., illustrating how the four key components operate together within the network [105].

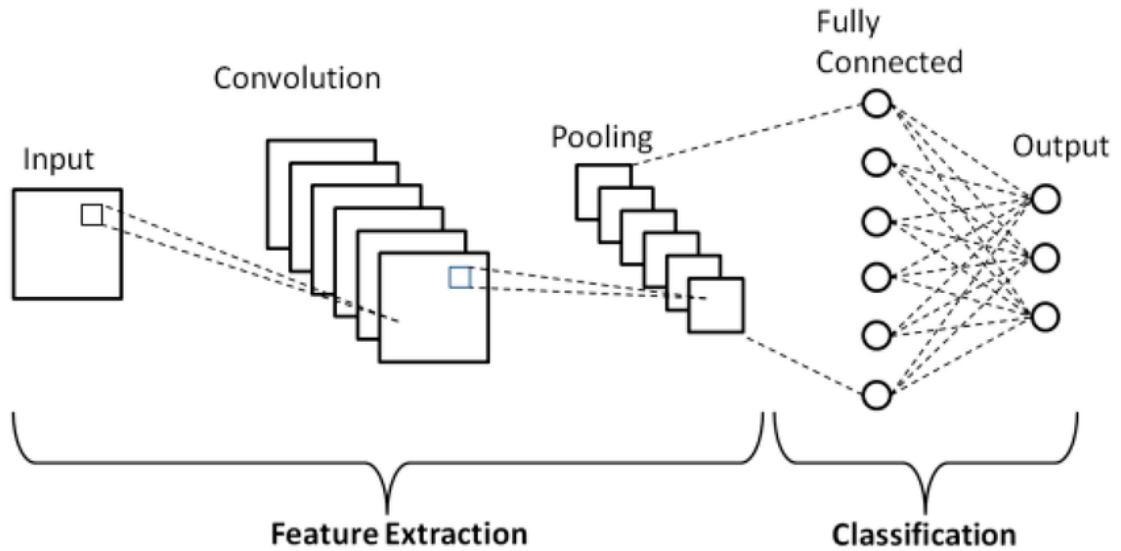


Figure 20. Schematic Diagram of Basic CNN Architecture.

CNNs have become foundational in modern deep learning applications due to their ability to automatically extract hierarchical features from data [106]. Unlike traditional deep learning algorithms which rely heavily on manual feature engineering, CNNs learn features directly from non-augmented input, significantly improving scalability and efficiency [102]. Their convolutional layers provide translation-invariant properties, enabling the model to detect patterns and features regardless of variations in position, orientation, scale, or translation [107]. Numerous pre-trained CNN architectures (e.g., VGG-16, ResNet50, InceptionV3, and EfficientNet) have achieved state-of-the-art results and can be fine-tuned for new tasks even with limited data [108][109]. Beyond image classification, CNNs demonstrate versatility across a wide range of domains, including natural language processing, time series analysis, and speech recognition [108].

2.4.2 CNNs in Medical Image Classification

Modern healthcare is increasingly reliant on advanced imaging techniques for accurate diagnosis and treatment planning [110]. Traditionally, skilled medical professionals have been responsible for analyzing complex medical images and

identifying subtle signs of disease [110]. However, limitations such as the scarcity of experts, fatigue-induced errors, and the time-consuming nature of manual feature extraction highlight the need for more efficient and robust technologies [111]. In this context, CNNs significantly advanced how visual data is processed and interpreted in the medical domain by enabling automated and accurate interpretation of visual data.

Wu et al. [112] presented a deep CNN model for breast cancer screening that was trained and evaluated on over one million medical images. The model, based on a custom ResNet architecture optimized for high-resolution data, demonstrated performance comparable to that of experienced radiologists. Its two-stage architecture and multi-view input design highlight how large-scale training and architectural innovations can yield clinically viable diagnostic tools.

A study by Stephen et al. [113] highlighted a CNN trained from scratch to classify and detect the presence of pneumonia. To address the challenge of limited pneumonia datasets, chest X-ray image samples were generated with several data augmentation algorithms to improve the validation and classification accuracy of the CNN model. This work underscores the importance of adapting CNN design and preprocessing techniques to address common constraints in medical imaging, such as limited and imbalanced datasets.

An analysis by Müller et al. [114] explored the performance impact of ensemble learning strategies in deep learning based medical image classification pipelines. The study investigated Augmenting, Stacking, and Bagging ensemble techniques, applying a pipeline with preprocessing and image augmentation methods. Results showed that Stacking yielded the largest performance gain with up to a 13% F1-score increase while Augmenting consistently improved performance by up to 4%. The authors concluded that integrating ensemble learning techniques can boost performance in medical image

classification. These results emphasize the importance of hybrid and ensemble approaches in improving the robustness and generalization of CNNs in clinical settings.

Despite their success, the widespread adoption of CNNs in clinical practice still faces challenges, including the limited availability of large annotated medical datasets, interpretability concerns, and ethical considerations [115]. In this thesis, CNNs are employed for their strong capability to extract high-level features from complex image data and to maintain classification performance even when faced with varying image quality.

2.4.3 Dense Convolutional Networks (DenseNet)

Proposed by Huang et al. [116], DenseNet was introduced to address issues like vanishing gradients, limited feature reuse, and weak feature propagation. DenseNet was created by combining the skip connection strategy from ResNet (Residual Neural Network) with the feature map concatenation approach from Inception. The core design principle of DenseNet is to ensure maximum information flow between layers by connecting each layer to every other layer in a feed-forward fashion. Unlike traditional CNNs that connect each layer only to its immediate successor (resulting in L connections for L layers), a DenseNet establishes $L(L + 1)/2$ direct connections, creating short paths from early layers to later layers, a concept taken from ResNet. This dense connectivity promotes efficient gradient flow and feature sharing across the network. Figure 21 taken from the study of Huang et al. illustrates the architecture of a deep DenseNet, visually outlining the core components and the steps involved in the model's forward flow. The architecture strengthens feature propagation, alleviates the vanishing gradient problem, encourages feature reuse, and reduces the number of parameters compared to traditional CNNs. DenseNet's advantages have led to its successful integration in a wide range of medical imaging tasks.

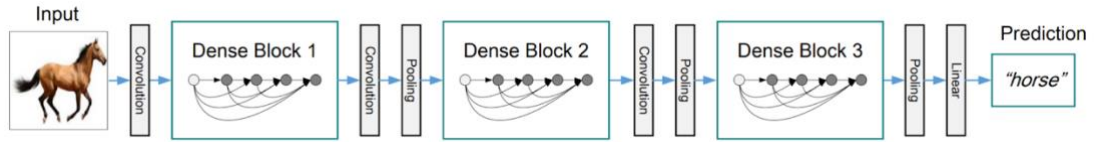


Figure 21. Sample of a Deep DenseNet using Three Blocks.

In a 2021 study, Hasan et al. employed DenseNet-121 for detecting COVID-19 from CT images to address limitations associated with traditional diagnostic methods, such as long processing times and reduced sensitivity to severe infections [117]. The model was trained on a dataset of 2,482 CT images and achieved an accuracy of 92% with a recall of 95%. DenseNet was selected due to its ability to mitigate the vanishing gradient problem, promote feature reuse, and reduce parameter redundancy, making it well-suited for deep medical image analysis. The use of image preprocessing and data augmentation further improved the model's robustness against overfitting, allowing for efficient and accurate diagnosis. This study demonstrates DenseNet's practical value in enabling rapid and automated COVID-19 detection in clinical environments.

DenseNet also serves as the backbone architecture in TRk-CNN, a model developed for glaucoma classification across three progressive stages: normal, glaucoma suspect, and glaucoma [118]. This task is particularly challenging due to the subtle, overlapping features associated with disease progression. DenseNet's densely connected layers allow each layer to receive inputs from all previous layers, improving information flow and enabling nuanced feature extraction. The TRk-CNN model achieved an overall accuracy of 92.96%, with sensitivity rates exceeding 93% for both glaucoma and glaucoma suspect categories. These results underscore DenseNet's effectiveness in complex diagnostic applications that require fine-grained differentiation between closely related medical conditions.

Another study proposed a deep learning framework for classifying brain tumors into four categories: gliomas, meningiomas, pituitary tumors, and non-tumor cases

[119]. The framework included adaptive filtering for noise reduction, Mask R-CNN for tumor segmentation, and GLCM for texture-based feature extraction. The model was trained on 2,870 images and tested on 394 images, achieving a classification accuracy of 96.96%, a sensitivity of 98.84%, and an F1-score of 97.70%. These results demonstrate the effectiveness of DenseNet-based architectures in supporting reliable, high-precision medical diagnoses from complex imaging data.

2.4.4 YOLOv11

Developed by the Ultralytics team, YOLOv11 is the latest version of the YOLO (You Only Look Once) series of object detection models [120]. This iteration improves performance, speed, and more efficient design, making it one of the most versatile options for object detection [121]. In the context of medical image analysis, particularly for tasks such as pressure ulcer classification where variable image quality and resource constraints are common, YOLOv11's enhanced performance and efficiency are highly relevant [122].

The architecture of YOLOv11 follows the core design philosophy of the YOLO family, which is unifying object localization and classification into a single end-to-end trainable framework. As illustrated in Figure 22, its architecture is composed of three main components: the backbone, neck, and head, each contributing to efficient feature extraction, aggregation, and prediction [120][123].

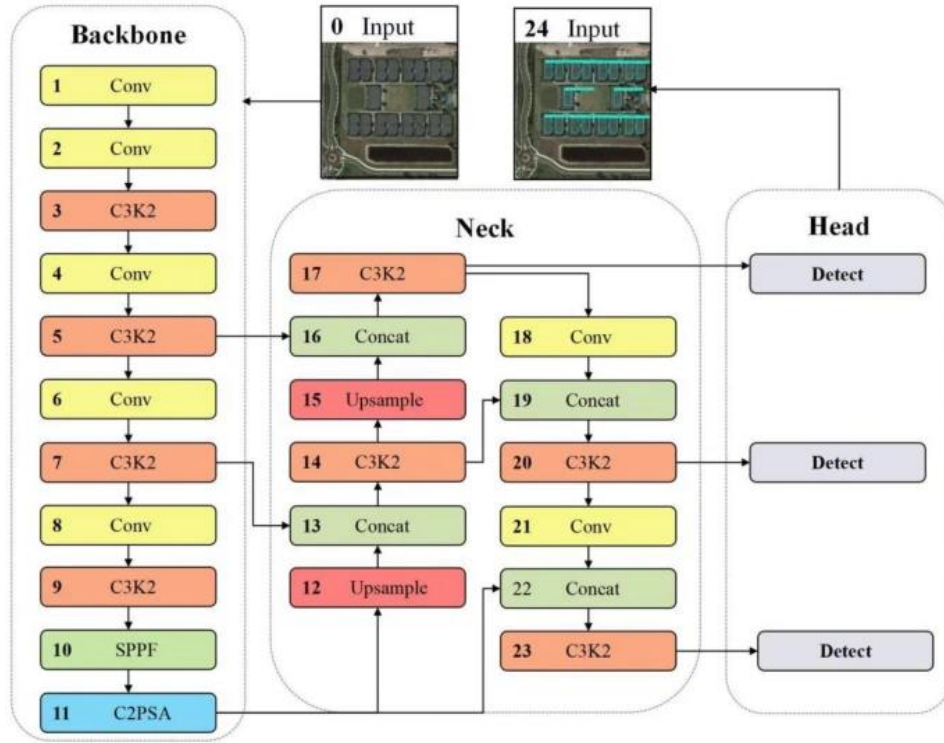


Figure 22. Structure Diagram of the YOLOv11 Network.

The backbone serves as the model's primary feature extractor and processes input images through successive convolutional (Conv) layers, followed by the C3k2, SPPF, and C2PSA modules [120][123]. The Conv module performs basic spatial feature extraction stabilized by batch normalization and the SiLU activation [120]. The C3k2 (Cross Stage Partial with kernel size 2) module introduces a multibranch structure with residual connections, designed to enhance feature richness while reducing computational cost [120][123]. Meanwhile, the SPPF (Spatial Pyramid Pooling - Fast) module aggregates multiscale contextual information via pyramid pooling, and the C2PSA (Convolutional block with Parallel Spatial Attention) module employs cross-scale pixel spatial attention to emphasize important image regions [120][123]. Together, these modules enable YOLOv11 to extract both fine-grained and global features efficiently [120].

The neck fuses and enhances the multi-scale features obtained from the backbone. Through upsampling and concatenation, it integrates detailed local patterns

with higher-level semantic information. The inclusion of C3K2 and C2PSA modules in this stage ensures that features are spatially attentive and computationally optimized, improving detection robustness for small or overlapping objects [123].

Finally, the head is responsible for the detection stage. It uses a multibranch design to perform bounding box regression and object classification simultaneously across different scales [120]. Each detection branch concludes with Conv2D and CBS (Convolution–BatchNorm–SiLU) layers, followed by the Detect module, which outputs bounding box coordinates, objectness scores, and class probabilities [120][123]. Non-Maximum Suppression (NMS) is applied to refine overlapping detections and yield the results [120].

Figure 23 compares YOLOv11 with earlier versions (from YOLOv5 to YOLOv10). The results show that YOLOv11 achieves higher accuracy (mAP) while maintaining faster inference speeds across all model sizes, with YOLOv11x reaching about 54.5% mAP at 13 ms latency [123].

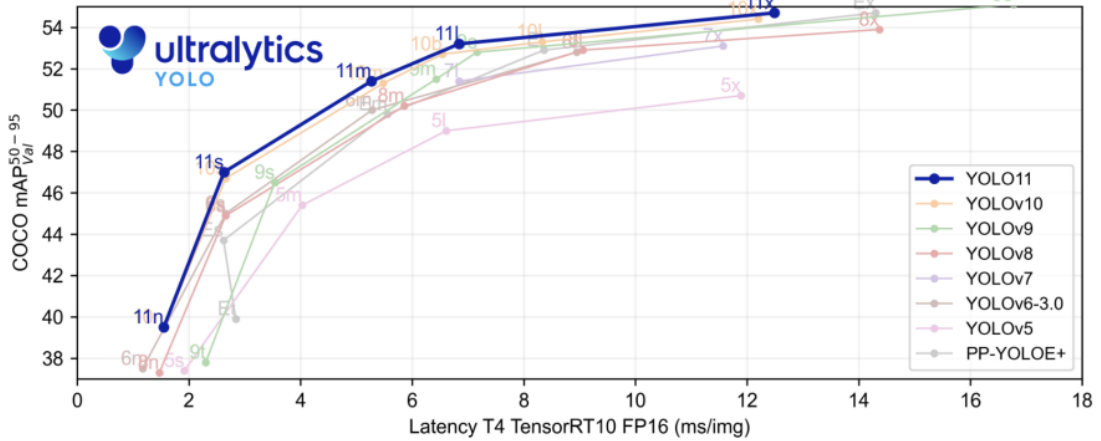


Figure 23. Benchmark of YOLOv11 Against Previous Versions.

YOLOv11 is well-suited for medical image analysis due to its ability to handle variable image quality, its need for speed, and its efficiency in low-resource environments [121]. The improved architecture and optimized training pipelines of

YOLOv11 deliver faster processing speeds while maintaining an optimal balance between accuracy and performance, making it suitable for deployment on devices with limited resources such as mobile phones [124]. In the context of pressure ulcer classification which involves identifying and staging ulcers from images that have varying lighting, resolution, and occlusion, YOLOv11's enhanced feature extraction and attention mechanisms are beneficial [124]. For instance, YOLOv11 has demonstrated superior performance in detecting smaller and more occluded objects compared to previous versions [125].

While DenseNet models, such as U-DenseNet, have shown impressive performance in medical image segmentation by effectively capturing fine anatomical details and promoting feature reuse, they are primarily designed for classification and segmentation tasks rather than real-time object detection [126]. DenseNet's dense connectivity helps mitigate the vanishing gradient problem and reduces the number of parameters while enhancing feature propagation. However, tasks requiring both detection and segmentation in real-time, especially with variable image quality and resource constraints, a YOLOv11-based approach offers distinct advantages. YOLOv11's ability to perform real-time object detection and instance segmentation in a unified framework, balancing speed and accuracy, makes it a promising solution for medical image analysis [14][127]. Studies have already explored the application of YOLOv11 for various medical imaging tasks, including brain tumor detection and segmentation, demonstrating its reliability and potential for clinical integration [128]. While DenseNet excels in detailed segmentation, YOLOv11's strengths lie in its real-time performance and efficiency for detecting and localizing objects within complex medical images, even those of mixed quality [121].

2.5 Gaps in Existing Studies

2.5.1 Base Reference of the Study

The study titled “Convolutional Neural Network Models for Visual Classification of Pressure Ulcer Stages: Cross-Sectional Study” investigates the application of convolutional neural networks (CNNs) in classifying pressure injury (PI) images into six stages: Stage I, Stage II, Stage III, Stage IV, unstageable, and suspected deep tissue injury (SDTI) [9]. The study utilized a dataset of 853 non-augmented PI images, which were augmented through cropping and flipping techniques to expand the dataset to 7,677 images. To assess the models’ performance, these images were then divided into training, validation, and test sets in an 8:1:1 ratio. AlexNet, VGGNet16, ResNet18, and DenseNet121 were the four CNN architectures that were put to the test. DenseNet121 outperformed the others in classifying the distinct stages of pressure ulcers, achieving the highest overall accuracy of 93.71%. While the other models performed commendably, their accuracies were lower: ResNet18 achieved 92.42%, VGGNet16 recorded 87.74%, and AlexNet reached 82.42%.

This study demonstrates the effectiveness of CNN-based models in the medical field, particularly for the classification of medical images, where accurate and automated pressure ulcer staging can improve patient care and clinical decision-making. The results highlight how crucial it is to choose models carefully and how important data augmentation methods are to get the best model performance. DenseNet121's high accuracy indicates that deeper networks with more intricate architectures can perform better than simpler models in tasks requiring fine-grained classification, like visualizing medical conditions.

The use of machine learning models for medical data analysis and classification is the focus of the study (especially in the wireless industry and its technological

advancements). This study offers important latest information about how CNNs can be used successfully for related healthcare tasks. The project explores different machine learning techniques to analyze and classify complex data, relates to the model evaluation, dataset preparation, and performance optimization principles described in the base paper. The success of DenseNet121 in this study motivates us to investigate deep learning models in our research, as these architectures may enhance the efficacy of predictive models in our field. Furthermore, machine learning models are ensured to be robust and generalizable by data augmentation and cross-validation techniques. These qualities can be important in a variety of datasets, like those we come across in wireless industry performance analysis.

2.5.2 Summary of Related Studies

To contextualize the development of automated pressure ulcer classification systems, this section presents a summary of related studies that have explored various deep learning-based approaches. The table below outlines the models, performance metrics, and summary from recent literature on pressure ulcer segmentation, categorization, and related wound types. These studies employed a range of architectures (from CNN-based pipelines like VGG16 and EfficientNet to transformer-based frameworks and multimodal systems) demonstrating the diversity of techniques used to address the challenges in wound image analysis. Performance evaluations include metrics such as accuracy, precision, recall, F1 score, and mean average precision (mAP), highlighting both the strengths and limitations of each approach.

	Study	Model(s)	Evaluation(s)	Notes
[9]	Lei et al. (2025)	DenseNet121	A = 0.9371 P = 0.9872 R = 0.9153 F1 = 0.9701	This study applied 4 different CNN architectures to do identify which model can efficiently and

				effectively assist in the accurate staging of pressure injuries.
[120]	Tsai (2025)	Yolov11n, FUSegNet	A = 0.99 PR = 0.909 F1 = 0.87 at 0.534	This study utilized a two-stage pipeline for wound segmentation, primarily aiming to support automated DESIGN-R® assessment.
[130]	Fergus et al. (2022)	Faster R-CNN	mAP = 0.7743	The neural network was designed to classify images of pressure ulcers into various categories and automatically categorize and report pressure ulcers in a standardized manner.
[131]	Shalom et al.	VGG16, ResNet152, EfficientNet-B2	Not explicitly stated, use a <i>multi-metric paradigm</i> .	The research utilized a multi-modal network based on a deep convolutional neural network for wound classification. It was adopted from the C-Net architecture. to categorize wounds into four specific types: diabetic, pressure, surgical, and venous ulcers
[132]	Zhong et al. (2024)	MSSDA	A = 0.889 P = 0.916 R = 0.926 F1 = 0.921	The research utilizes a novel deep learning framework called MSSDA (Multi-Sub-Source Adaptation). This framework is not a single, monolithic model but rather a three-stage pipeline that integrates various deep learning components and strategies to recognize diabetic foot neuropathy.

[133]	Mousa et al. (2025)	ViT, DWT, Transformer	A = 0.8123	This study utilized a novel deep learning architecture called the multimodal classifier to address the challenge of classifying acute and difficult-to-heal wounds into various categories, including diabetic, pressure, surgical, and venous ulcers.
[134]	Bloch et al. (2021)	EfficientNet, pix2pixHD	Not provided	The study used a hybrid model to classify the infection and ischaemia in Diabetic Foot Ulcer (DCU) images.
Accuracy = A Precision = P Recall = R F1 Score = F1				

Table 1. Summary of Related Studies on Pressure Ulcer Classification and Evaluation Using Deep Learning Approaches

The base reference for this thesis mentioned in chapter 2.5.1 is summarized in the last row of Table 1. The study by Lei et al. [9] demonstrated the effectiveness of DenseNet121 in accurately classifying pressure ulcer stages using CNN architectures and image augmentation techniques. Building upon this foundation, the present study expands the methodological approach by incorporating DDPMs and StyleGAN2-ADA, aiming to improve performance on varied image datasets.

Chapter 3

METHODOLOGY

This chapter presents the methods and procedures employed to develop and evaluate deep learning models for classifying pressure ulcer severity across four stages (I–IV). The study conducts eight experimental runs that combine different data settings, ranging from purely real (NON-AUGMENTED) images to hybrid datasets augmented with synthetically generated samples from the Early-Stopped Denoising Diffusion Probabilistic Model (ES-DDPM) and StyleGAN2-ADA. These generative models are used to address data scarcity and improve model generalization. Guided by the conceptual framework, this chapter details each methodological phase, including data gathering, preprocessing, generative data synthesis, model architecture design, training, validation, and evaluation.

3.1 Conceptual Framework

The conceptual framework outlines the structured approach of this study to evaluate eight model runs by crossing four data settings; non-augmented, ES-DDPM, StyleGAN2-ADA, and ES-DDPM + StyleGAN2-ADA, with two architectures; DenseNet and YOLOv11 to classify pressure ulcers into Stages I-IV. The diagram shown in Figure 24 provides a visualization of the pipeline: Data gathering, data preprocessing, data setting and data splitting, classification, learning (training, validation, evaluation), and comparative analysis.

3.1.1 Data Gathering

The data gathering phase involves collecting diverse pressure ulcer images from multiple sources to ensure clinical relevance, quality, and variation in visual characteristics. The finalized dataset consists of 2,001 images categorized into four

clinical stages: Stage I (n = 499), Stage II (n = 509), Stage III (n = 517), and Stage IV (n = 476).

To support consistent training and unbiased model evaluation, the dataset was divided into three subsets following a 70: 15: 15 ratios. The distribution across classes is summarized below:

- **Training Set** (1,395 images, $\approx 69.7\%$): Stage I = 343, Stage II = 345, Stage III = 353, Stage IV = 354
- **Validation Set** (303 images, $\approx 15.1\%$): Stage I = 77, Stage II = 81, Stage III = 88, Stage IV = 57
- **Testing Set** (303 images, $\approx 15.1\%$): Stage I = 79, Stage II = 83, Stage III = 76, Stage IV = 65

This stratified split ensures proportional representation of all stages in each subset, maintaining class balance, and preventing sampling bias during model training and evaluation.

Images were obtained from three primary sources:

1. **Hospital and Nursing Home Settings:** Real-world cases were gathered through Mapúa University's School of Nursing department, leveraging its network of healthcare professionals and institutions. These images capture various severity levels and conditions of pressure ulcers, contributing to the dataset's clinical reliability.
2. **Nursing Education Materials:** Standardized images from medical literature were incorporated to provide reference-quality examples for each ulcer stage. These materials support consistency and serve as benchmarks for model validation.

3. Publicly Available Online Repositories: Reliable online medical databases and open-access resources were used to expand the dataset volume and enhance variability in image conditions and backgrounds.

All collected images were strictly anonymized and contained no demographic or patient-identifiable information. Only wound-focused photographs were used, excluding any faces, tags, timestamps, or metadata that could identify individuals. As the dataset involves no personal health information, official ethical clearance was not required. Mapúa University's School of Nursing department oversees the data collection and verification process to ensure ethical compliance and data integrity.

3.1.2 Data Preprocessing

Preprocessing is intentionally minimal and consistent across runs. Each image is resized to 224 x 224 pixels to match the input dimensions required by CNNs, and to maintain consistency across samples. The pixel values are normalized to range between [0, 1] to improve model stability and convergence during training. Additionally, horizontal, and vertical flips with probability of 0.5 are applied as data augmentation. No other augmentations such as rotation, shear, zoom, or color jitter were introduced. Stage labels (I-IV) are assigned to all images for supervised learning, and the dataset is then split into training, validation, and test sets. Each image in the dataset is annotated using labels with its corresponding pressure ulcer stage (I-IV). For the annotation process, we engage with an expert to ensure that the labeled data is accurate and consistent. Expert input is vital for medical image datasets because proper identification of pressure ulcer stages is required to train the model effectively. This consultation helps refine the classification criteria and address any ambiguities that may occur during the labeling process, ensuring that the dataset is representative and accurately categorized.

For the model runs that use ES-DDPM and StyleGAN2-ADA to expand training data, the generated images are excluded from the non-augmented baseline and undergo the same preprocessing pipeline before training. Through these pre-processing procedures, the dataset becomes uniform, allowing the models to focus on prominent features rather than irrelevant visual inconsistencies.

3.1.3 Augmentation Setting and Data Splitting

After preprocessing, four data settings are defined wherein they differ only in whether synthetic images are added to the training material:

1. Non-Augmented (baseline): The training, validation, and test sets contain only the original 2,013 images; no synthetic data is used.
2. ES-DDPM: The training and validation sets are expanded with ES-DDPM samples; the test set remains non-augmented.
3. StyleGAN2-ADA: The training and validation sets are expanded with StyleGAN2-ADA samples; the test set remains non-augmented.
4. ES-DDPM + StyleGAN2-ADA (combo): The training and validation sets are expanded with outputs from both generators; the test remains non-augmented.

Labeled datasets from each data setting are then aligned to a single stratified split defined on the non-augmented dataset with a fixed random seed: Training (70%), Validation (15%), and Test (15%). The test split is created once and contains non-augmented images only; it is reused across all eight runs for unbiased evaluation. In the generative settings, ES-DDPM and StyleGAN2-ADA samples are generated after splitting and merged exclusively into the training and validation sets. The same split indices are shared by DenseNet and YOLOv11.

The four data settings crossed with the two architectures yield eight runs:

Data Setting	Stage 1	Stage 2	Stage 3	Stage 4	Total
Non-Augmented	350/74/75	357/76/76	371/79/79	334/71/71	2014
ES-DDPM	420/89/89	427/91/91	432/92/92	404/86/86	2401
StyleGAN2-ADA	420/89/89	427/92/92	432/92/92	404/86/86	2401
ES-DDPM + StyleGAN2-ADA	490/104/104	497/106/106	493/105/105	474/101/101	2789

Table 2. Eight Model Runs.

3.1.4 Classification (DenseNet and YOLOv11)

The classification stage is where the prepared data is processed by the selected models to generate a prediction. In this phase, the training and validation sets from each of the four distinct data settings (1. Non-Augmented, 2. ES-DDPM, 3. StyleGAN2-ADA, and 4. ES-DDPM+StyleGAN2-ADA) paired with the two chosen CNN model architectures: DenseNet and YOLOv11. This systematic pairing establishes the eight distinct experimental runs for the study. From this point, the paired models and datasets are passed to the “Learning Process Phase” (Chapter 3.1.5), where they will be actively trained, validated, and evaluated to measure their performance. The key steps in the process, from input reception to output generation, are as follows:

1. DenseNet

- **Input Reception:** DenseNet receives an image resized to 224×224 pixels from the preprocessing pipeline. To ensure consistent input scaling across all images, the pixel values are normalized to a range between [0, 1].
- **Feature Extraction and Processing:** DenseNet utilizes its dense block architecture, where each layer receives input from all previous layers. It can extract complex and hierarchical features at several layers due to this feature reuse. Convolutional layers

gradually extract low-level to high-level features, including edges, textures, and patterns, as the image passes through the network. This hierarchical feature extraction helps DenseNet learn fine-grained details in medical images. While bottleneck convolutions lower dimensionality to avoid overfitting, batch normalization and ReLU activation functions stabilize the learning process.

- **Output Generation:** The final features are processed by a fully connected layer with a softmax activation after going through several dense blocks and pooling layers. The probability distribution for each of the four ulcer stages is produced by the softmax function. By choosing the class with the highest probability, the model predicts the pressure ulcer stage.

2. YOLOv11

- **Input Reception:** Like DenseNet, YOLOv11 also receives a 224×224-pixel image after preprocessing, with pixel values normalized to [0, 1]. However, YOLOv11's approach focuses more on localization and detection of key regions rather than pixel-level classification.
- **Feature Extraction and Processing:** The three components of YOLOv11's architecture are the Head, Neck, and Backbone.
 - **Backbone:** The backbone creates feature maps by extracting hierarchical features from the unprocessed image, such as edges and textures.

- Neck: To analyze and combine features with different spatial resolutions, the neck aggregates and refines multi-scale feature maps from the backbone.
- Head: Using a softmax activation, this last layer combines the features and generates the probability distribution for each of the four stages.

The Neck helps YOLOv11 handle features at different scales, making it efficient for both large and small regions of interest, which is important for detecting ulcers that may vary in size.

- Output Generation: Each region of interest (in this case, the entire image) receives a categorization prediction from YOLOv11, with the ulcer stage having the highest predicted probability. As YOLOv11 is designed for real-time inference, it is more computationally efficient for real-time applications that require quick predictions.

3.1.5 Learning Process Phase (Training, Validation, Evaluation)

The learning process involves three phases: Firstly, the models are trained on the training split and within each model family (DenseNet and YOLOv11), the training schedule and hyperparameters are kept constant across all four data settings (Non-Augmented, ES-DDPM, StyleGAN2-ADA, ES-DDPM + StyleGAN2-ADA) to isolate the effect of the data setting. Secondly, the validation split is used for early stopping and checkpoint selection to help prevent overfitting and promote generalization. Lastly, for evaluation, the final performance is computed on the fixed non-augmented test split by all eight runs. Per run, the confusion matrix, overall accuracy, macro-precision, and macro-recall are reported. Because the same test set and evaluation protocol are used

for every run, observed differences can be attributed to the data setting and model family, not to evaluation variability.

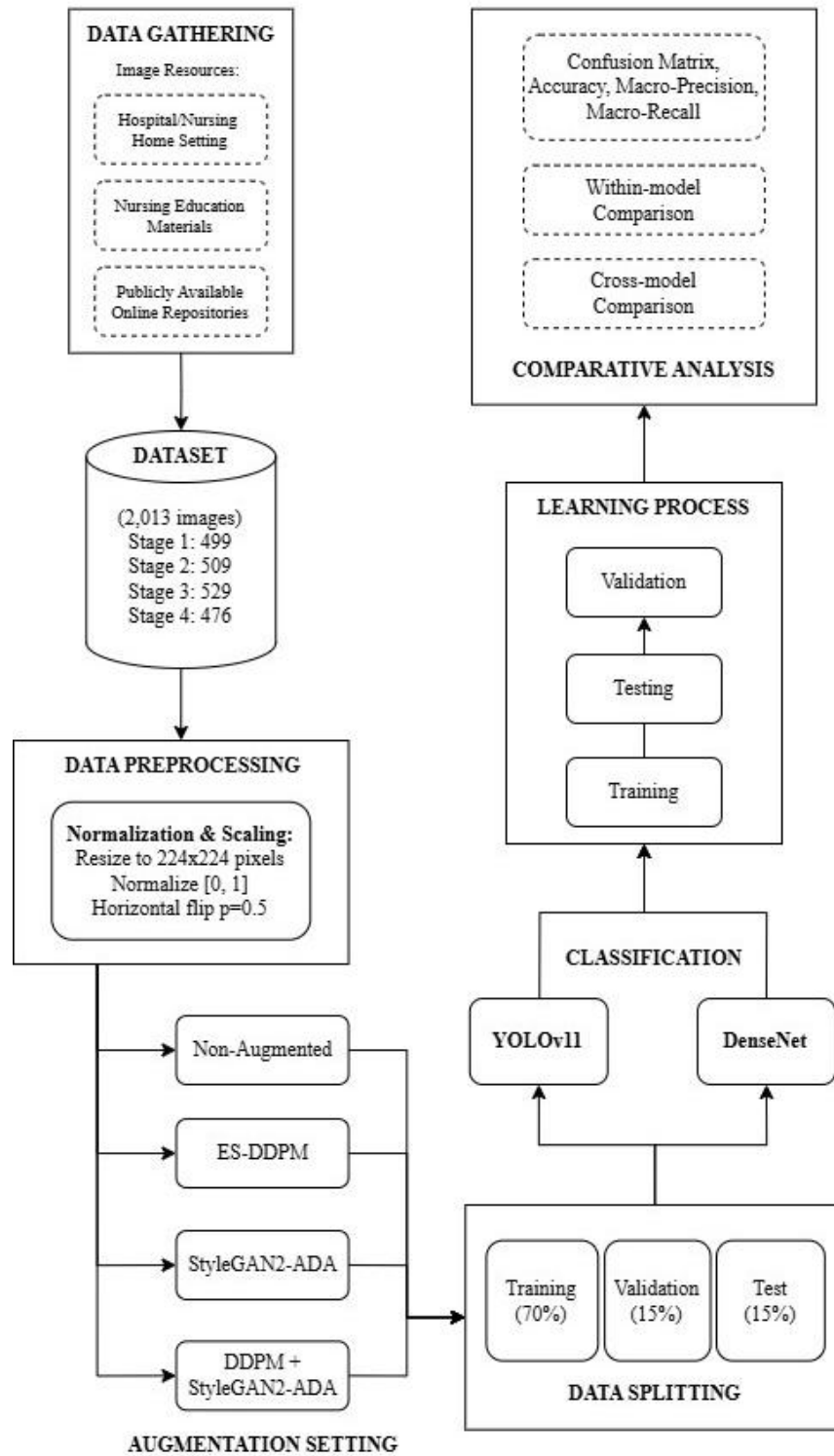


Figure 24. Conceptual Framework.

3.2 Generative Data Synthesis

3.2.1 Early Stopped DDPM

The ES-DDPM uses a diffusion process to produce high-quality synthetic data. Diffusion and denoising are the two primary phases of this process. After learning to convert a clean image into a noisy one, the model reverses the process to recover the original image. The early-stopped DDPM's primary innovation is its ability to optimize the trade-off between computational efficiency and image quality by stopping earlier denoising.

1. **Diffusion Process:** The process starts by introducing noise to the image, gradually degrading the image x_0 (the clean input) into a noisy version x_T through a series of T steps. Each step adds more noise to the image, and by the time it reaches the final step x_T , the image has become indistinguishable from random noise. This is done using a predefined noise schedule, and the model learns how to predict the noise at each step.
2. **Denoising Process:** The model performs the denoising process after the image has been completely diffused (i.e., transformed into noise). It begins with x_T and aims to reverse the noise process, progressively recovering the clean image x_0 . The denoising is done by learning a conditional probability $p_\phi(x_{t-1}|x_t)$ to predict the noise added at each step, effectively removing the noise and refining the image back to its original state.
3. **Early Stopping:** One of the key features of this model is the early stopping technique, where the denoising process is terminated before reaching the final step $t = 0$. Stopping earlier (at some step T') helps

avoid overfitting and reduces the computational burden of performing full denoising. This results in a model that balances high-quality outputs with reduced computational costs, making it suitable for real-time applications. The latent code z is derived through encoding x_0 with a learned function $q_\psi(z|x_0)$. This code represents the information from the partially diffused image and guides the model during the denoising steps to reconstruct the original, clean image efficiently. The early stopping helps prevent overfitting by reducing the complexity of the diffusion process, resulting in faster inference and less computational cost while maintaining or improving the sample quality.

- 4. Process Overview:** The model includes an encoding and decoding process, where the clean image is first encoded into a latent code z . The decoder then works on the latent code to sample and produce the clean image. The denoising process is performed from the noisy image x_T back towards x_0 , but it is terminated early, optimizing the balance between image quality and computational cost.

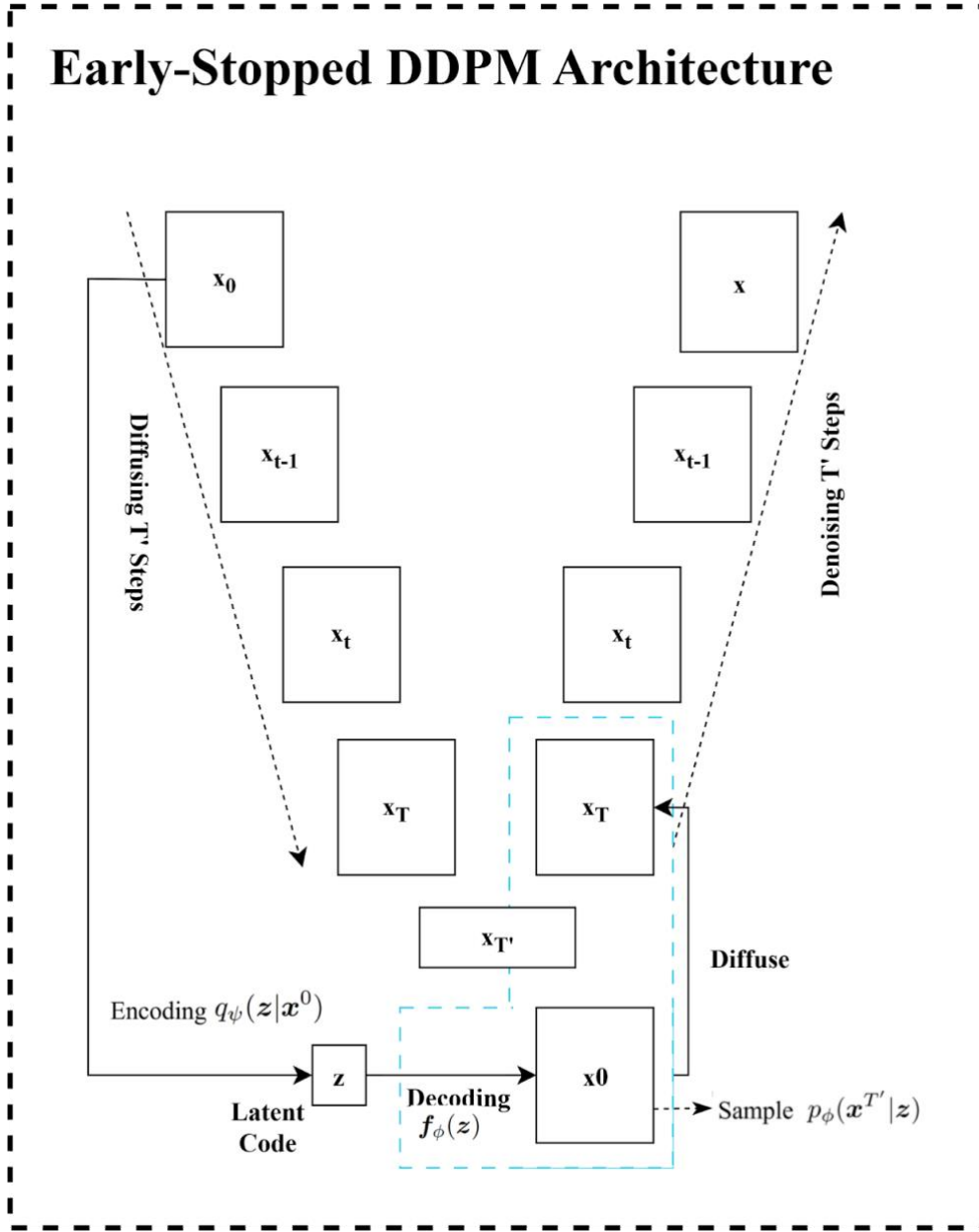


Figure 25. Early Stopped DDPM Architecture.

Early stopping speeds up the generation process and helps avoid the overfitting that may occur when the denoising continues too long. This is aligned with recent advancements in generative models, where diffusion processes have been shown to generate high-quality synthetic data that rivals other generative models like GANs (Generative Adversarial Networks) [7].

3.2.2 *StyleGAN2-ADA*

The StyleGAN2-ADA architecture generates high-quality synthetic images by using a generator network that learns to map latent vectors to image distributions. The model starts with a latent vector $z \in Z$, which undergoes normalization before being processed through fully connected layers. The transformed latent vector $w \in W$ is then passed through a series of operations involving AdaIN (Adaptive Instance Normalization) and convolutional layers to generate the synthetic images. The generator network consists of multiple layers where noise is introduced to enhance variation and produce high-quality output images [135].

The Synthesis Network processes the latent vector and synthesizes the image through AdaIN and convolutional layers. The network structure includes up-sampling and convolutional layers to refine the image progressively. The key innovation in StyleGAN2-ADA is using adaptive noise at various stages of the generation process to improve image quality and robustness against mode collapse, a common issue in generative adversarial networks (GANs) [135].

The model training involves comparing the generated synthetic dataset with the ground truth training dataset. The loss function is based on image quality, distribution matching, and the ground truth data, ensuring that the generator produces realistic and high-fidelity images. This architecture emphasizes robust training techniques, such as Adaptive Discriminator Augmentation (ADA), which helps the model generalize better, particularly in scenarios with limited data.

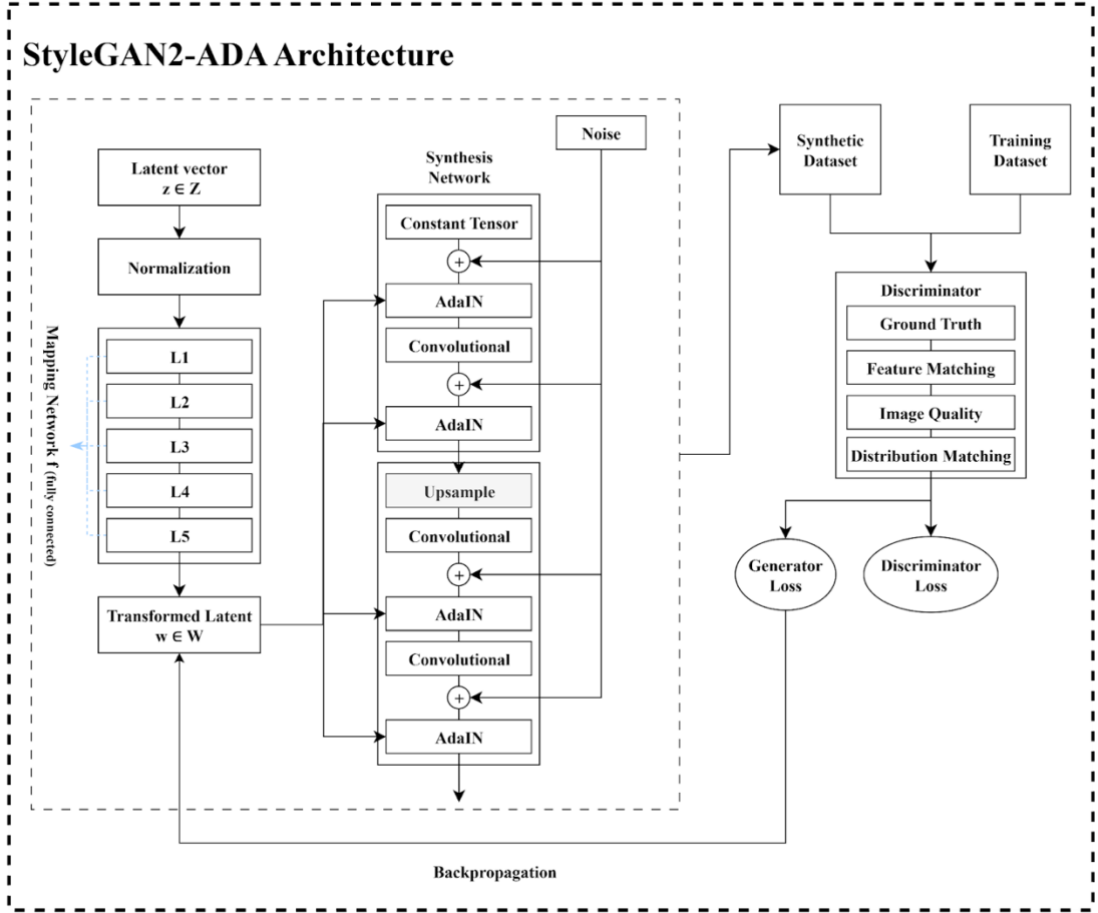


Figure 26. StyleGAN2-ADA Architecture.

3.3 Model Architectures

3.3.1 DenseNet

The DenseNet Architecture used in this study is a convolutional neural network (CNN) that leverages dense connections between layers, enabling more efficient feature reuse and improving gradient flow during training. This model consists of the following:

1. **Input Layer:** The input layer receives the non-augmented image data, which in the context of this study could be related to skin images used for classification (e.g., for pressure ulcer detection or other dermatological conditions). The image undergoes an initial

convolutional operation to extract basic features from the non-augmented input.

- 2. Dense Block:** The dense block holds the feature map, which consists of multiple layers that extract low-level features from the input. Each layer in this block is connected to all previous layers, meaning that the output of each layer includes not only the features from the current layer but also those from the earlier layers. This block uses several sub-components, including Batch Normalization, which helps stabilize the learning process and improves the model's generalization. ReLU Activation introduces non-linearity to the model. Bottleneck Convolution (2D), which reduces dimensionality to avoid overfitting. Convolution 2D extracts deeper features as the image passes through the block. Then it will be passed to the concatenation layer, ensuring that all the extracted features are preserved and flow into the subsequent layers. This dense connectivity allows the network to efficiently reuse features efficiently, resulting in better performance with fewer parameters.
- 3. Pooling:** After every process of the dense block, the concatenated feature map will pass through pooling to reduce the spatial dimensions and focus on the most critical features before transferring to the next block.
- 4. Prediction Layer:** After passing through all four dense blocks, the output is passed through a pooling layer, followed by a fully connected layer (linear), which produces the final prediction for multiple stages (stages I, II, III, and IV) in the context of pressure

ulcer detection. This allows the model to predict the severity of the condition at various stages.

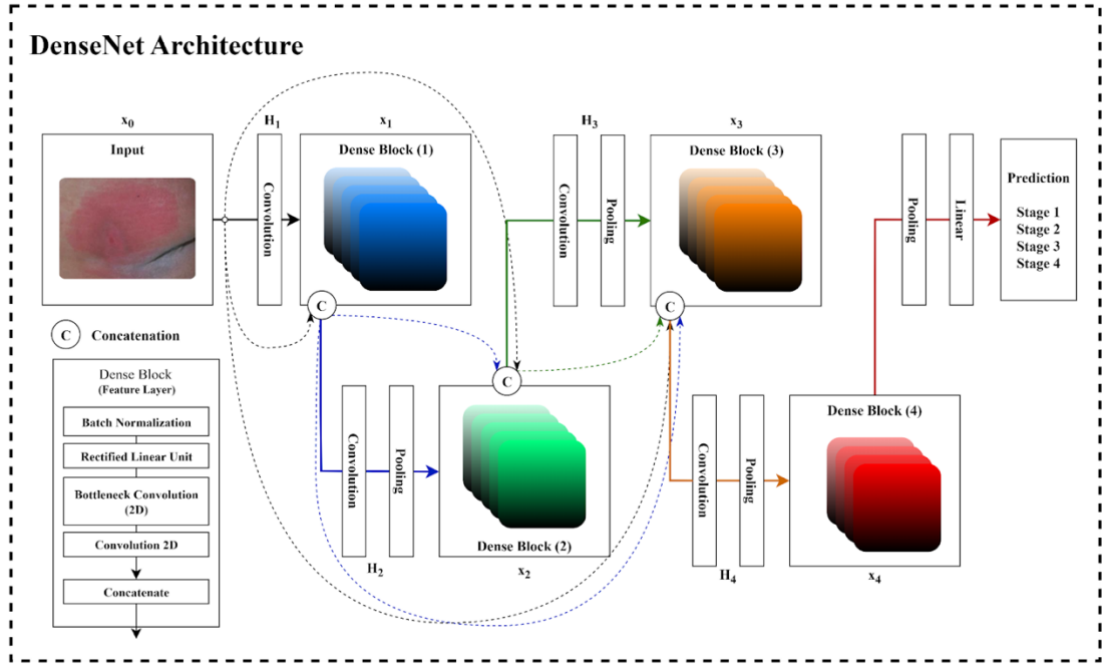


Figure 27. DenseNet Architecture.

DenseNet's unique feature is its use of dense connections, where each layer receives input from all previous layers in the network, making the network highly efficient and helping to mitigate the vanishing gradient problem that can occur in traditional CNNs. This model is developed to have satisfactory performance with fewer parameters while also being able to reuse features, leading to improved generalization capabilities and better performance in tasks like classification and detection.

3.3.2 YOLOv11

The YOLOv11s is a lightweight variant of the YOLO architecture, specifically optimized for image classification tasks that require rapid inference speeds and minimal computational resources. It employs a streamlined three-part structure consisting of a Backbone, Neck, and Head visually represented in Figure 28.

1. **Backbone:** This serves as the foundational feature extraction mechanism, responsible for transforming non-augmented input

images into hierarchical feature representations. This progressively extracts features ranging from low-level patterns (edges, textures) to high-level semantic representations, creating an increasingly abstract understanding of the image essential for classification.

2. **Neck Network:** This functions as a feature aggregation and refinement module, synthesizing the multi-scale feature maps generated by the backbone. This ensures the model can recognize visual information regardless of its size in the image, retaining only essential aggregation that has a balance between computational speed and effective multi-scale feature integration.
3. **Classification Head:** This is the final decision-making module, transforming aggregated features into classification predictions. This layer summarizes the presence of learned features across the entire image while reducing dimensionality. This compact feature is then processed by one or more fully connected layers, culminating in a final layer that uses a SoftMax activation function to output a probability distribution across all predefined classes.

The approach involves adapting the model from object detection to a purely classification task by removing detection-specific components like bounding box regression and objectness prediction. This adaptation provides several crucial advantages: it significantly improves computational efficiency for faster inference, simplifies the training process by focusing on a single classification objective, and allows for resource optimization, making the lightweight design ideal for deployment on resource-constrained devices. The model learns task-specific features, such as helmet shape, color, or texture, through end-to-end supervised training.

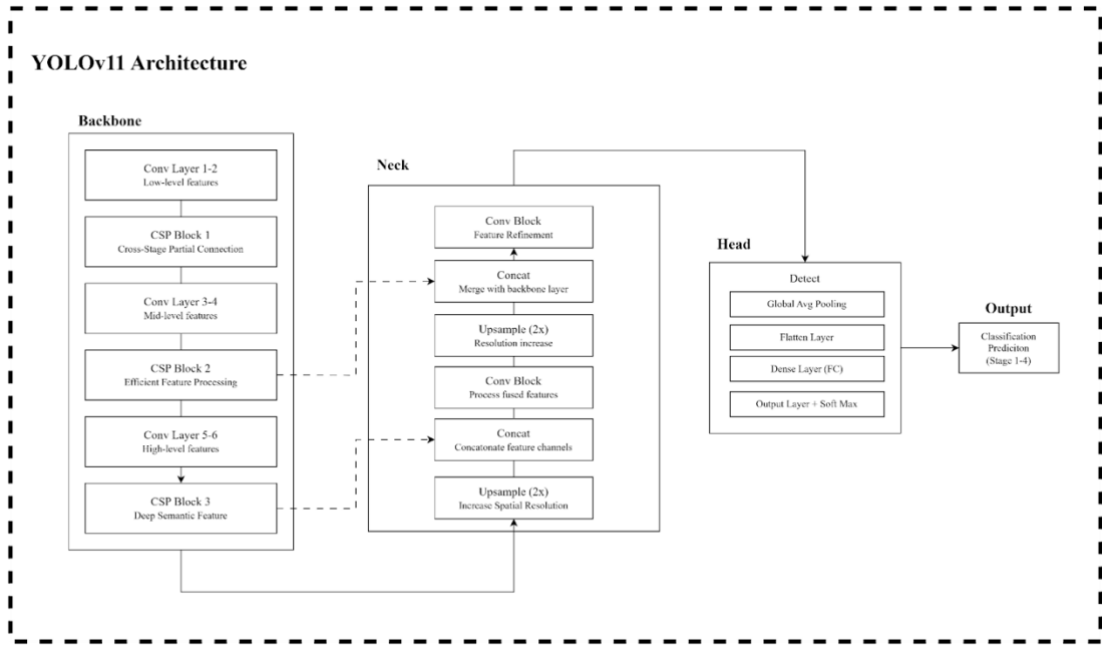


Figure 28. YOLOv11 Architecture.

3.4 Training Setup

The training setup displays the exact training configurations used across all models to ensure reproducibility and fair comparison. Common settings across the two data augmentation models and the two CNN models are aligned where possible, while model-specific controls are stated explicitly.

ES-DDPM		
Hyperparameter	Value	Description
img_size	256	Squares resize (center-crop → bicubic resize to $\text{IMG_SIZE} \times \text{IMG_SIZE}$).
channels	3	Image channels (RGB).
batch_size	4	Training batch size.
num_gpus	1	Number of GPUs used for training.
epochs	20	Number of epochs over the dataset.
accum_steps	2	Gradient accumulation steps (<i>effective batch = batch_size × accum_steps</i>).
workers	0-2	Data-loader worker threads.

optimizer	AdamW(lr=2e-4)	Optimizer for network parameters.
grad_clip	1.0	Gradient-norm clipping to stabilize training.
t_total	1000	Total diffusion timesteps are used during training.
beta_schedule	“cosine”	Cosine schedule for β during diffusion.
net	TinyUNet (c = 3, base_ch = 32)	Lightweight U-Net for ε -prediction.
base_ch	32	Base channel width of the U-Net.
norm/act	GroupNorm(8) + SiLU	Normalization and activation are used in convolution blocks.
time_embedding	MLP: 1→128→128	Embeds timestep t_total for conditioning.
crop/resize	center-crop → bicubic resize	Make square crops then resize to img_size.
scaling	[-1, 1]	Pixel normalization to [-1, 1] (and channel move to CHW).
loader	shuffle = True, drop_last = True	Randomized mini batches; Drop incomplete batch.
sampling_steps	50 (100 - 250 for high quality)	Steps for DDIM-like fast sampler.

Table 3. ES-DDPM Hyperparameters.

Table 3 details the hyperparameters used for the ES-DDPM model, configured for efficient training on 256×256 RGB images. The setup utilizes a lightweight TinyUNet architecture trained on a single GPU, employing gradient accumulation to manage effective batch sizes over 20 epochs. The diffusion process is governed by a cosine schedule across 1000 timesteps, while inference uses a faster sampling method to reduce computational costs during generation.

StyleGAN2-ADA		
Hyperparameter	Value	Description
img_size	256	Image resolution (height/width).

batch_size	8	Training batch size.
num_gpus	1	Number of GPUs used for training.
gamma	8	Regularization strength; controls smoothness of generated images
learning_rate	0.002	Learning rate for training the model.
truncation_psi	0.7	Controls the variability of generated images.
z_dim	512	Dimensionality of the latent vector.
fmap_base	8192	Base number of filters in the first convolutional layer.
fmap_decay	0.5	Controls the decay of the number of filters per layer in the network.
fmap_max	512	Maximum number of filters in any layer.
d_lrate	0.002	Learning rate for the discriminator during training.
g_lrate	0.002	Learning rate for the generator during training.
r1_gamma	10.0	Regularization of strength for the discriminator’s R1 loss helps prevent overfitting.

Table 4. StyleGAN2-ADA Hyperparameters.

Table 4 outlines the configuration for the StyleGAN2-ADA model, which was trained on 256×256 images using a single GPU and a batch size of 8. The model’s generator and discriminator were trained with equal learning rates (0.002) and stabilized using R1 regularization. Key architectural settings include a 512-dimensional latent space (z_dim), and generated samples use a truncation_psi of 0.7 to balance image fidelity and diversity.

Hyperparameter	ES-DDPM	StyleGAN2-ADA
Image size	256	256
Batch size	4	8
GPUs used	1	1
Learning rate	AdamW(lr = 2e-4)	g_lrate = 0.002, d_lrate = 0.002

Regularization	grad_clip = 1.0	gamma = 8, r1_gamma = 10.0
Augmentation	crop→resize; scaling to [-1, 1]	aug = noah, mirror = 1
Model capacity knob	base_ch = 32	fmap_base = 8192, fmap_decay = 0.5, fmap_max = 512
Generation control	sampling_steps = 50	truncation_psi = 0.7

Table 5. ES-DDPM and StyleGAN2-ADA Similar Hyperparameters.

Table 5 compares the shared hyperparameters for ES-DDPM and StyleGAN2-ADA, revealing different approaches to the same task. Both models were configured to generate 256x256 images on a single GPU. However, StyleGAN2-ADA used a larger batch size (8 vs. 4) and a significantly higher learning rate (0.002 vs. 2e-4) with a different optimization scheme. The two models also employed fundamentally different regularization methods: ES-DDPM used gradient clipping, while StyleGAN2-ADA relied on gamma and R1 regularization. Finally, they use distinct parameters to control model complexity (base_ch vs. fmap_base) and to manage the output during inference (sampling_steps vs. truncation_psi).

DenseNet		
Hyperparameter	Value	Description
base_model	DenseNet121	DenseNet pre-trained on ImageNet.
num_layers	121	121 layers for DenseNet.
imgsz	224	Input image size used for train/val.
output_layer	softmax activation	For multi-class classification.
optimizer	Adam	Optimizers are used for training.
learning_rate	1e-4	Base learning rate for Adam.

training_augmentation	Horizontal flipping, Rescaling [0, 1]	Horizontal flipping and pixel values rescaled to [0, 1] for the training set.
validation_augmentation	Rescaling [0, 1]	Rescaling only, no augmentation
epochs	20	Number of training epochs.
batch_size	32	Batch size for generators and fitting.

Table 6. DenseNet Hyperparameters.

Table 6 outlines the hyperparameters for the DenseNet121 model, which was fine-tuned for multi-class classification using 224×224 inputs. Training was conducted over 20 epochs with a batch size of 32, utilizing the Adam optimizer (lr=1e-4). Data processing involved standard rescaling for all images, with horizontal flipping added during the training phase.

YOLOv11s		
Hyperparameter	Value	Description
base_model	YOLOv11s (classification)	YOLOv11 small variant in classification mode, ImageNet-pretrained.
model_weights	yolo11s-cls.pt	Pretrained weights.
num_layers	86 (train graph); 47 (fused)	Layers reported by Ultralytics summary during training/inference.
imgsz	224	Input image size used for train/val.
output_layer	Classifier head (softmax at inference)	Linear logits trained with cross-entropy; softmax used at inference for multi-class outputs.
optimizer	AdamW (auto)	Selected by optimizer=auto.
learning_rate	1.25e-3 (auto)	Base LR chosen by Ultralytics with AdamW

training_augmentation	RandAugment (auto_augment), resize to 224	Ultralytics classification pipeline with automatic RandAugment; standard normalization
validation_augmentation	Resize/normalize to 224	No extra augments for validation.
epochs	20	Number of full passes over the training set.
batch	32	Batch size per iteration during training.
workers	2	DataLoader workers
val_split	0.2 (80/20) when missing	If no validation images exist, 20% of training images are moved to validation.

Table 7. YOLOv11s Hyperparameters.

Table 7 summarizes the configuration for the YOLOv11s classification model, which uses ImageNet-pretrained weights and an input size of 224×224. The model was trained for 20 epochs with a batch size of 32, using an automatically selected AdamW optimizer ($\text{lr}=1.25\text{e-}3$). Training incorporated automated RandAugment for data augmentation, while validation relied solely on standard resizing and normalization.

Hyperparameter	DenseNet	YOLOv11s
base_model	DenseNet121 (ImageNet)	YOLOv11s (classification, ImageNet-pretrained)
num_layers	121	86 (train graph); 47 (fused)
imgsz	224	224
output_layer	softmax activation	classifier head (softmax at inference)
optimizer	Adam	AdamW (auto)
learning_rate	1e-4	1.25e-3 (auto)
training_augmentation	Horizontal flipping, Rescaling [0, 1]	RandAugment (auto_augment), resize to 224; standard normalization

validation_augmentation	Rescaling [0, 1]	Resize/normalize to 224
epochs	20	20
batch_size	32	32

Table 8. DenseNet121 and YOLOv11s Similar Hyperparameters.

Table 8 compares the hyperparameters for DenseNet and YOLOv11s, showing identical foundation settings: both used 224×224 inputs, a batch size of 32, and ran for 20 epochs using ImageNet pre-trained weights. The primary differences lie in their training pipelines, where DenseNet used a standard Adam optimizer with simple horizontal flipping, while YOLOv11s utilized an automated AdamW setup with advanced RandAugment for data augmentation.

3.5 Model Validation

The model validation phase involves assessing the performance of each trained model on the validation set, which contains images that were not used during training. This process evaluates the model's generalization to unseen data and helps identify potential issues like overfitting or underfitting. The validation set includes both authentic pressure ulcer images and, for generative runs, synthetically augmented images produced by ES-DDPM and StyleGAN2-ADA, ensuring diverse representation of image conditions.

During validation, key performance metrics such as accuracy, precision, recall, and loss trends are monitored across training epochs. These metrics guide the fine-tuning of hyperparameters (e.g., learning rate, batch size, and regularization strength) to optimize model generalization. Visualization of training and validation curves is used to observe convergence behavior, detect early signs of overfitting, and determine whether the model has reached an optimal balance between bias and variance.

Given the variability in pressure ulcer image quality arising from differences in lighting, angle, and texture, validation plays a critical role in ensuring that the model

learns clinically meaningful features rather than memorizing training samples. When performance gaps between training and validation sets are observed, architectural parameters or data preprocessing settings are adjusted to stabilize learning. The best-performing model checkpoint, determined by the highest validation accuracy and lowest loss, is then saved and carried forward to the evaluation phase for final testing.

3.6 Evaluation Protocol & Metrics

In this study, the researchers evaluated the models' performance using a range of evaluation indicators. Various metrics were selected according to the task, such as object classification for DenseNet and YOLOv11 and image processing for StyleGAN2 and ES-DDPM.

3.6.1 Evaluation Metrics for Augmentation

To quantitatively assess the performance of the augmentation models ES-DDPM and StyleGAN2-ADA, the following standard metrics are used [136]:

- **FID (Fréchet Inception Distance)** value is a measure of the discrepancy between the distribution of the generated image and the real image. A lower FID value indicates a higher quality of the produced image. It is denoted by the formula:

$$FID = \|\mu_r - \mu_g\|^2 + Tr(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}})$$

- **KID (Kernel Inception Distance)** is like FID but uses an unbiased estimator of the squared Maximum Mean Discrepancy (MMD) between the distributions of real and generated images. It is less sensitive to the number of images used, which is especially helpful in evaluating image quality when data is scarce. It is denoted by the formula:

$$KID(x, y) \|\mathbf{m}_x - \mathbf{m}_y\|^2 + Tr(\mathbf{C}_x + \mathbf{C}_y - 2(\mathbf{C}_x \mathbf{C}_y)^{\frac{1}{2}})$$

- **LPIPS (Learned Perceptual Image Patch Similarity) Diversity Score** computes the perceptual similarity between images. It uses deep learning models to compare features extracted from the images to assess their perceptual similarity. A low LPIPS score indicates that the images are perceptually similar.
- **IS (Inception Score)** evaluates the quality and diversity of images generated by AI models. It works by using a pre-trained Inception model to assess how realistic and varied the generated images are. A high score indicates that the images are sharp, recognizable, and cover a wide range of classes. It is denoted by the abbreviated formula:

$$IS(G) = \exp (E_{x \sim p_g} D_{KL}(p(y|x) || p(y)))$$

3.6.2 Evaluation Metrics for Classification

To quantitatively assess classification performance of DenseNet and YOLOv11, the models are evaluated using standard metrics:

- **Accuracy** refers to the percentage of total predictions that the model correctly classifies. It is denoted by the formula:

$$accuracy = \frac{true\ positives + true\ negatives}{true\ positives + true\ negatives + false\ negatives + false\ positives}$$

- **Precision** measures the proportion of correctly predicted positive instances out of all cases the model labels as positive, calculated as:

$$precision = \frac{true\ positives}{true\ positives + false\ positives}$$

- **Recall (Sensitivity)** evaluates the model's ability to identify all actual positive cases, defined by the formula:

$$recall = \frac{true\ positives}{true\ positives + false\ negatives}$$

- **F1-Score** is the harmonic mean of precision and recall and serves as a single metric that balances both concerns, denoted by the formula:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

- **Confusion Matrix** is used to evaluate the performance of a model and is visually represented as a table. It tallies true (rows) vs. predicted (columns) labels; diagonal cells show correct predictions while diagonals show confusion.
- **Support** for each class refers to the number of actual instances or samples from the dataset that belong to that specific class. It provides a count of each class's actual occurrences in the test dataset, indicating how frequently it appears in the dataset under evaluation.

In this study, Precision and Recall were used as key evaluation metrics for both image augmentation and classification processes. These metrics were chosen to provide a comprehensive view of model performance in both image generation and classification tasks.

Chapter 4

RESULTS AND DISCUSSION

This chapter presents the results of the different models used to classify pressure ulcer stages (1-4). The models tested include combinations of non-augmented data with DenseNet and YOLO, as well as generative models like DDPM and StyleGAN paired with DenseNet and YOLO. The purpose of this chapter is to evaluate how well each model performs in classifying the ulcer stages, and to examine the impact of using generative models on the classification process. By comparing these models, the study aims to understand whether the inclusion of generative models improves the ability of convolutional neural networks (CNNs) to accurately classify the various stages of pressure ulcers.

4.1 NON-AUGMENTED Data Setting

In this section, the study examines the performance of convolutional neural networks (CNNs) when applied to non-augmented image data for classifying pressure ulcer stages. The non-augmented data approach focuses on using the images without any additional preprocessing or augmentation techniques, allowing the CNN models to learn directly from the non-augmented features in the images. By evaluating these models, the study aims to understand how well they can extract meaningful patterns and accurately classify the distinct stages of pressure ulcers based on unprocessed visual information.

4.1.1 NON-AUGMENTED + DenseNet

The core performance metrics for the non-augmented data combined with the DenseNet model are shown in the table below.

Macro Averages	
Precision	83.85%
Recall	84.60%

F1-score	84.13%
Weighted Averages	
Precision	83.58%
Recall	83.83%
F1-score	83.61%
Overall Accuracy	
Accuracy	83.83%

Table 9. The model achieved an overall accuracy of 83.83%. The macro averages indicate an overall impressive performance of the model, with precision at 83.85%, recall at 84.60%, and an F1-score of 84.13%. These values suggest that the DenseNet model, when trained on non-augmented data, is capable of reliably identifying pressure ulcer stages with a balance between false positives and false negatives. The weighted averages, which take class imbalances into account, show similarly strong metrics with precision at 83.58%, recall at 83.83%, and F1-score at 83.61%. This highlights the model's effectiveness across all ulcer stages, despite variations in class distribution.

Macro Averages	
Precision	83.85%
Recall	84.60%
F1-score	84.13%
Weighted Averages	
Precision	83.58%
Recall	83.83%
F1-score	83.61%
Overall Accuracy	
Accuracy	83.83%

Table 9. NON-AUGMENTED + DenseNet: Core Performance.

The per-class metrics shown in Table 10 and illustrated in Figure 29 provide further insight into the model's performance on each pressure ulcer stage. The precision and recall for Stage I are strongest, at 89% precision and 97% recall ($F1 = 93$, $n = 77$), indicating reliable detection of early pressure ulcer formation. Stage II drops to 82% precision and 77% recall ($F1 = 79$, $n = 81$), suggesting more borderline misses and false

negatives. Stage III is the weakest, with 80% precision and 75% recall ($F1 = 77$, $n = 88$), pointing to confusion with adjacent stages. In contrast, Stage IV rebounds to 86% precision and 89% recall ($F1 = 87$) despite the smallest support ($n = 57$), implying that severe ulcers are more visually distinctive. Overall, errors cluster around the mid-stage boundary (II-III) rather than the extremes (I and IV).

Class	Precision	Recall	F1-Score	Support
Stage I	89%	97%	93%	77
Stage II	82%	77%	79%	81
Stage III	80%	75%	77%	88
Stage IV	85%	89%	87%	57

Table 10. NON-AUGMENTED + DenseNet: Per-Class Metrics.

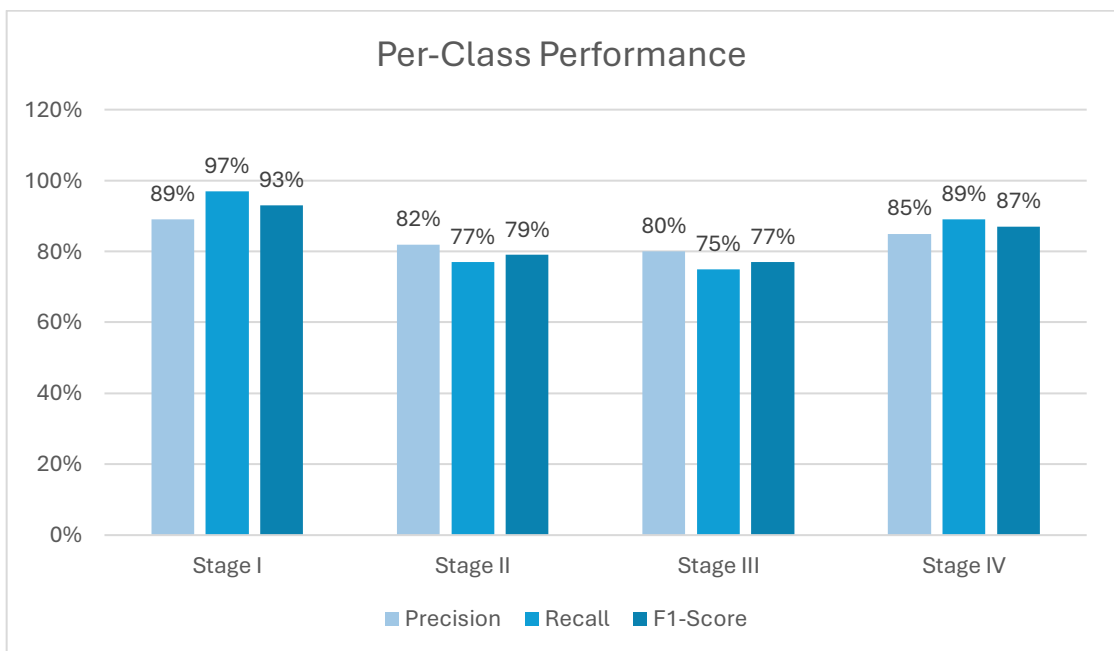


Figure 29 NON-AUGMENTED + DenseNet Per-Class Performance Bar Chart.

The confusion matrix shown in Figure 30 further complements the performance metrics, providing a visual representation of the model's classification results. Stage I remains in the most reliably identified class, with 75 correct predictions and only 2 samples misclassified as Stage II. However, Stage II shows notable spillover to

neighboring classes with 62 correct predictions, but 8 instances predicted as Stage I and 11 as Stage III, indicating boundary ambiguity. Stage III presents the most prominent cross-stage confusions with only 66 correct predictions but 22 total misclassifications (12 cases labeled as Stage II, 9 as Stage IV, and 1 as Stage I), highlighting challenges in separating adjacent severity levels. Stage IV is well recognized (51 correct), though 6 samples are misclassified as Stage III, reflecting residual overlap at the severe end. Overall, the diagonal entries (75, 62, 66, 51) indicate solid class recognition, while the off-diagonal counts highlight that the most difficult distinctions lie between adjacent stages; particularly II↔III and III↔IV.

True Label	Stage I	Stage II	Stage III	Stage IV	
	75	2	0	0	
	8	62	11	0	
	1	12	66	9	
	0	0	6	51	
		Predicted Label			
	Stage I	Stage II	Stage III	Stage IV	

Figure 30. NON-AUGMENTED + DenseNet: Confusion Matrix.

Overall, the NON-AUGMENTED + DenseNet model demonstrates strong classification capabilities, achieving an overall accuracy of 83.83% with balanced macro and weighted metrics across standard performance indicators. Contrary to a simple linear decline in performance with increasing severity, the model excels at identifying the extremes; Stage I is detected most reliably, and Stage IV shows a notable rebound in performance, suggesting that severe ulcers remain visually distinctive despite smaller sample sizes. The primary challenge for this model lies in mid-stage

boundary ambiguity, as Stages II and III exhibit the weakest individual metrics and the highest rates of inter-class confusion. While the model shows significant promise for automated classification, further refinements, potentially through generative data augmentation, are necessary to better resolve the subtle visual overlaps between these intermediate stages.

4.1.2 NON-AUGMENTED + YOLOv11

Table 11 presents the core performance metrics for the YOLOv11 model trained on non-augmented data, showing an overall accuracy of 88.44%. The model demonstrates robust capabilities, evidenced by its macro averages: 88.83% precision, 89.17% recall, and an F1-score of 88.98%. These figures indicate a strong balance in the model's ability to correctly identify positive cases while limiting false positives. Crucially, the weighted averages, which account for class imbalances, remain high (precision at 88.44%, recall at 88.45%, and F1-score at 88.43%), confirming that YOLOv11 maintains its effectiveness across all ulcer stages despite differing sample sizes.

Macro Averages	
Precision	88.83%
Recall	89.17%
F1-score	88.98%
Weighted Averages	
Precision	88.44%
Recall	88.45%
F1-score	88.43%
Overall Accuracy	
Accuracy	88.44%

Table 11. NON-AUGMENTED + YOLOv11s: Core Performance.

The per-class evaluation shown in Table 12 and Figure 33 provides more detailed insights into the performance for each ulcer stage. I achieved the highest performance, with 98.68% precision, 97.40% recall, and 98.04% F1-score, indicating excellent identification of early-stage ulcers. Stage IV also performed well, achieving

a 92.31% F1 score. In contrast, Stage II and Stage III displayed lower F1 scores of 83.75% and 81.82%, respectively. These results suggest that YOLOv11 tends to perform best in identifying the more visually distinct ulcer stages (Stage I and Stage IV) but struggles slightly with intermediate stages where the visual boundaries and texture differences are subtler. The decrease in recall for Stage II and III may indicate partial confusion between these adjacent stages.

Class	Precision	Recall	F1-Score	Support
Stage I	98.68%	97.40%	98.04%	77
Stage II	84.81%	82.72%	83.75%	81
Stage III	81.82%	81.82%	81.82%	88
Stage IV	90.00%	94.74%	92.31%	57

Table 12. NON-AUGMENTED + YOLOv11s: Per-Class Metrics.

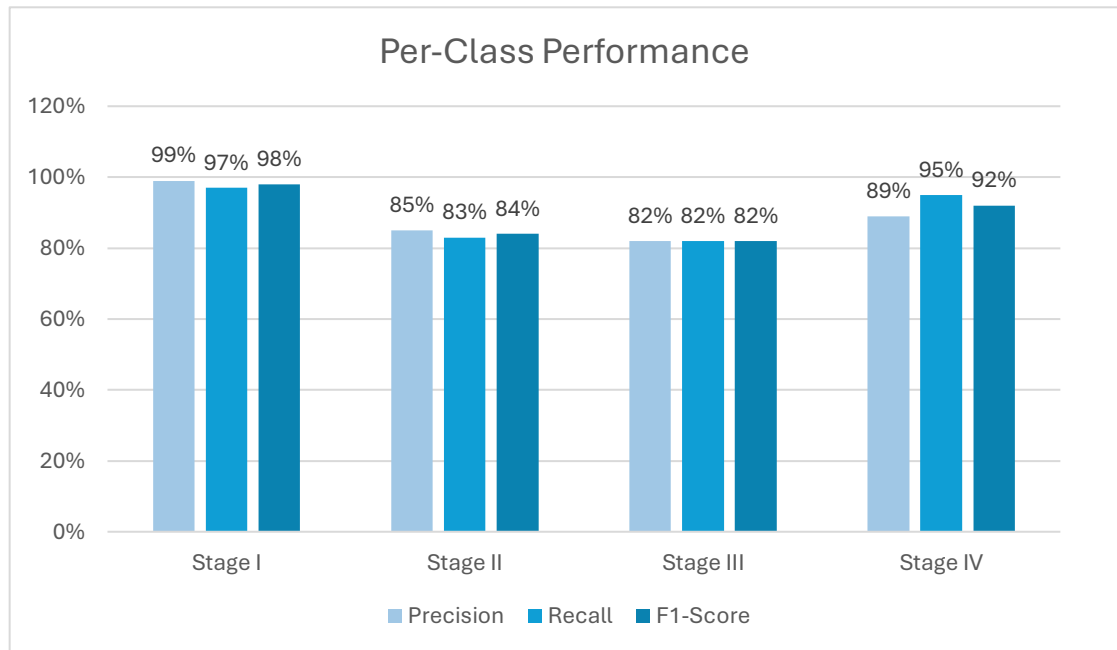


Figure 31. NON-AUGMENTED + YOLOv11s Per-Class Performance Bar Chart.

The confusion matrix shown in Figure 32 further details the classification performance of YOLOv11, highlighting specific areas of strength and confusion. Stage I detection is reliable, with 75 (out of 77) true samples correctly identified and only 2

misclassified as Stage II. The model's primary challenge lies in the ambiguity between adjacent stages with the most prominent confusion occurring between Stage II and Stage III, where 13 Stage II samples were mislabeled as Stage III, and 10 Stage III samples were mislabeled as Stage II. Confusion is also seen between Stage III and Stage IV, with 6 Stage III cases labeled as Stage IV and 3 Stage IV cases labeled as Stage III. Despite these boundary challenges, performance at the extremes is strong: Stage I is robustly identified, and Stage IV recognition is also high, with 54 out of 57 images correctly classified.

True Label	Stage I	Stage II	Stage III	Stage IV
	75	2	0	0
	1	67	12	0
	0	10	72	6
Stage IV	0	0	3	54
		Predicted Label		
		Stage I	Stage II	Stage III

Figure 32. NON-AUGMENTED + YOLOv11s: Confusion Matrix.

Overall, the YOLOv11 model trained on non-augmented data demonstrates strong classification performance, achieving an accuracy of 88.44%. Its results suggest that YOLOv11 can efficiently process and classify wound images without relying on additional preprocessing or generative augmentation. Compared to the DenseNet model, YOLOv11 shows a notable improvement in overall metrics, though it shares a similar challenge in differentiating between the intermediate stages (Stage II and Stage III), as seen in the confusion matrix. Despite this mid-stage ambiguity, YOLOv11 proves to be a reliable model for real-time ulcer stage classification, excelling at identifying early

(Stage I) and late (Stage IV) cases and showing its strong potential for integration into automated wound assessment systems.

From the results of the non-augmented + DenseNet and Non-augmented + YOLOv11 models, it can be observed that YOLOv11 outperforms DenseNet across most evaluation metrics. The YOLOv11 model achieved an accuracy of 88.45%, while DenseNet reached 83.83%, indicating that YOLOv11 was more effective in extracting relevant features from non-augmented wound images. While both models faced challenges in distinguishing intermediate ulcer stages (Stage II and III), YOLOv11 demonstrated higher overall consistency and stronger performance at the extremes (Stage I and Stage IV). This suggests that YOLOv11's modern architecture is well-suited for this classification task even without generative augmentation. Consequently, YOLOv11 not only offers superior precision but also shows greater promise for efficient, real-time medical image classification compared to the DenseNet baseline.

4.1.3 Baseline Model Comparative Analysis

The analysis of per-class metrics demonstrates that the classification performance varied among the four pressure ulcer stages, which signifies that the dataset's visual characteristics played a significant role in the model's accuracy. Both models' F1 scores were the highest on Stage I (DenseNet: 93%, YOLOv11s: 98.04%) and Stage IV (DenseNet: 87%, YOLOv11s: 92.31%). The reason for this was that the classes were not ambiguous as Stage I was characterized by a change in color on intact skin, while Stage IV was identified by severe tissue loss. The unique visual characteristics of these two allow both models to share the same features and easily extract and classify them from the images.

On the other hand, both DenseNet and YOLOv11s had their lowest F1 scores in Stages II and III. The problem is that the two stages are similar visually, and the difference between partial-thickness loss (Stage II) and shallow full-thickness loss (Stage III) is subtle. The confusion matrices (Table 11 and Table 14) point to this as the main failure, showing a substantial portion of misclassification between Stages II and III in both models. This systematic confusion pattern that is not network-dependent strongly indicates that the main issue is a data itself, one stemming from the visual ambiguity of the dataset rather than a specific architectural weakness.

Despite that fact, all models were dealing with the same underlying data problem; the YOLOv11s model easily surpassed DenseNet in terms of F1-score for all four classes. This is because the difference in their architectural designs was the root cause of the discrepancy in performance. DenseNet is a classification network that looks at features globally throughout the whole image whereas YOLOv11s is a detection network that focuses on specific parts of the image to find objects.

4.2 ES-DDPM Data Setting

This section presents the results of applying the Early-Stopped Denoising Diffusion Probabilistic Model (ES-DDPM) to the pressure ulcer dataset. The ES-DDPM approach introduces controlled Gaussian noise to the original images and then performs a partial denoising or reverse process, allowing the model to generate enhanced and more diverse representations of the data. As shown in Figure 33, each image undergoes a transition from its original state to a noisy version and finally to a reversed form that retains essential structural features while reducing noise. This process aims to improve data quality and variability, providing a more robust dataset for CNN-based classification. By generating these representations, the ES-DDPM model enhances the network's ability to recognize subtle visual differences between

ulcer stages, potentially leading to better classification performance in the succeeding experiments.

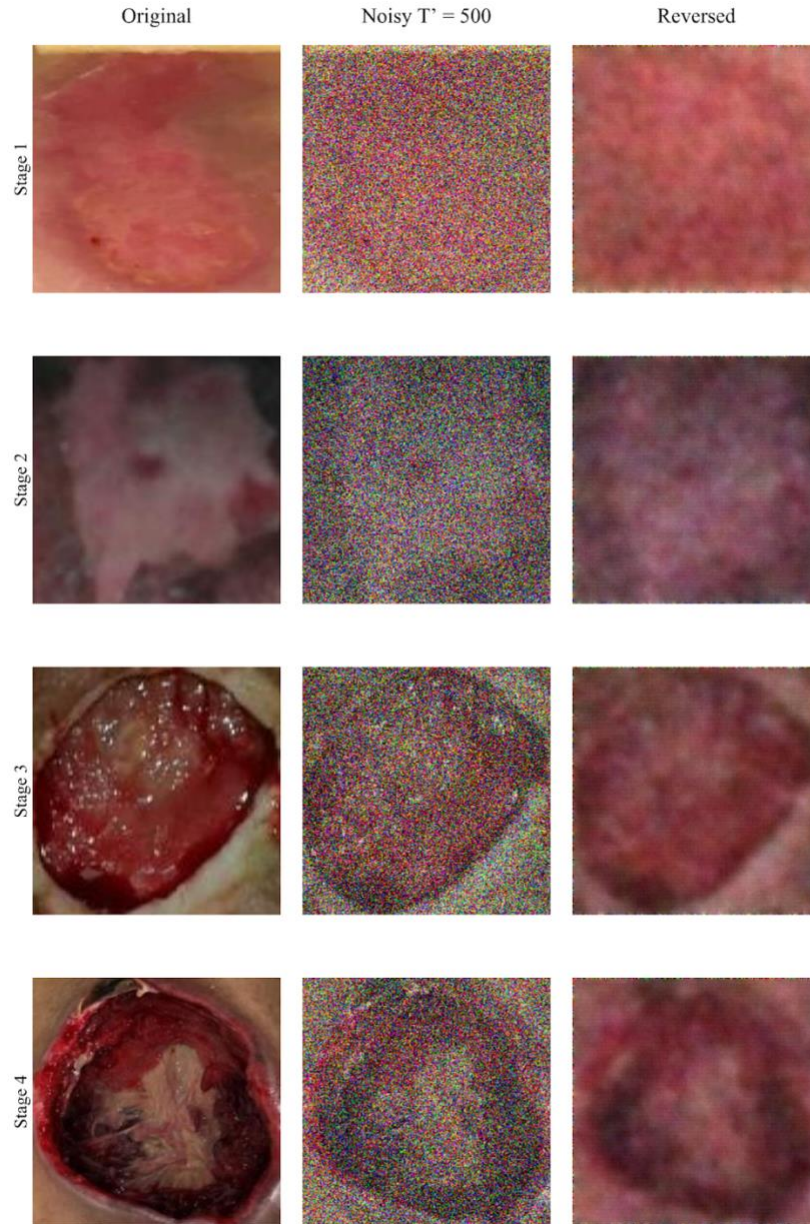


Figure 33. ES-DDPM Original and Reversed Images.

The stage-specific evaluation results show how the Early-Stopped Denoising Diffusion Probabilistic Model (ES-DDPM) performed in generating images for each pressure ulcer stage. For Stage I, the model achieved an Inception Score (IS) of 3.48 ± 0.15 , suggesting moderate quality and diversity among generated samples. However, the high FID of 394.38 and KID of 0.4399 ± 0.0014 indicate that the generated images

still differ noticeably from the real ones. The LPIPS diversity score of 0.2050 ± 0.0109 implies limited variation among samples, though sufficient to capture the basic texture patterns of early-stage ulcers.

Stage I	
Metric	Score
Inception Score (IS)	3.48 ± 0.15
Fréchet Inception Distance (FID)	394.38
Kernel Inception Distance (KID)	0.4399 ± 0.0014
LPIPS Diversity Score	0.2050 ± 0.0109
Number of Pairs (n_pairs)	500

Table 13. ES-DDPM Stage I Metrics.

For Stage II, the Inception Score improved to 4.12 ± 0.58 , showing higher visual quality and better-defined ulcer representations. Despite this, the FID increased to 441.35, and KID rose to 0.5183 ± 0.0014 , implying that while the generated images became clearer, they drifted slightly further from the real data distribution. The LPIPS score of 0.2273 ± 0.0131 shows a small gain in diversity, reflecting the greater variability in Stage II wound appearances.

Stage II	
Metric	Score
Inception Score (IS)	4.12 ± 0.58
Fréchet Inception Distance (FID)	441.35
Kernel Inception Distance (KID)	0.5183 ± 0.0014
LPIPS Diversity Score	0.2273 ± 0.0131
Number of Pairs (n_pairs)	500

Table 14. ES-DDPM Stage II Metrics.

The Stage III results yielded the highest Inception Score (4.51 ± 0.42), suggesting that the model generated more realistic and visually coherent samples for this stage. However, the FID (530.31) and KID (0.6962 ± 0.0021) values also peaked, signaling a trade-off between improved perceptual quality and statistical similarity to real data. The LPIPS diversity remained consistent at 0.2073 ± 0.0111 , indicating that while Stage III outputs were visually sharp, their overall variation was still moderate.

Stage III	
Metric	Score
Inception Score (IS)	4.51 ± 0.42
Fréchet Inception Distance (FID)	530.31
Kernel Inception Distance (KID)	0.6962 ± 0.0021
LPIPS Diversity Score	0.2073 ± 0.0111
Number of Pairs (n_pairs)	500

Table 15. ES-DDPM Stage III Metrics.

For Stage IV, the model attained an Inception Score of 3.80 ± 0.32 , with FID = 529.00 and KID = 0.6987 ± 0.0019 , showing a slight decline in image quality and realism compared to Stage III. Nonetheless, the LPIPS diversity reached 0.2415 ± 0.0140 , the highest among all stages, suggesting improved sample variation for severe ulcer images, due to the more complex textures and color gradients found in advanced stages.

Stage IV	
Metric	Score
Inception Score (IS)	3.80 ± 0.32
Fréchet Inception Distance (FID)	529.00
Kernel Inception Distance (KID)	0.6987 ± 0.0019
LPIPS Diversity Score	0.2415 ± 0.0140
Number of Pairs (n_pairs)	499

Table 16. ES-DDPM Stage IV Metrics.

Overall, the ES-DDPM results reveal a consistent pattern where later ulcer stages (Stage III and IV) tend to produce higher Inception Scores but also higher FID and KID values. This indicates that the model captures more distinctive visual details as ulcer severity increases but struggles to maintain perfect alignment with the real data distribution. The moderate LPIPS diversity across all stages confirms that the early-stopped diffusion approach preserved key visual structures while introducing a controlled level of variability. In summary, the ES-DDPM effectively generated

realistic and stage-specific synthetic data that can enhance CNN-based classification performance in the succeeding model evaluations.

4.2.1 ES-DDPM + DenseNet

The results of the ES-DDPM + DenseNet model shown in Table 15 indicate an overall accuracy of 86.11%, with macro averages of 85.93% precision, 85.86% recall, and an F1-score of 85.86%. These metrics suggest that the model performs consistently across all ulcer stages, without showing significant bias toward any class. The weighted averages are similarly high, with 86.16% precision, 86.11% recall, and an 86.11% F1-score, further confirming balanced classification. While the performance shows a slight improvement over the baseline DenseNet model, the high uniformity of scores reflects that the generative data produced by the ES-DDPM model contributed to better generalization and dataset diversity.

Macro Averages	
Precision	85.93%
Recall	85.86%
F1-score	85.86%
Weighted Averages	
Precision	86.16%
Recall	86.11%
F1-score	86.11%
Accuracy	
86.11%	

Table 17. ES-DDPM + DenseNet: Core Performance.

The per-class results shown in Table 18 and Figure 34 reveal noticeable variations in the model's ability to classify different ulcer stages. I achieved the highest performance, with 90% precision, 94% recall, and a 92% F1-score, showing the model's strong capability in detecting early-stage ulcers. Stage II also performed well, with an F1-score of 87% (both precision and recall at 87%), indicating stable recognition of mild ulcer features. However, Stage III recorded the lowest performance, with 76% precision and 77% recall, resulting in an F1-score of 77%, suggesting

persistent difficulty in distinguishing this stage due to its intermediate visual complexity.

In contrast, Stage IV achieved solid results, scoring 91% precision, 86% recall, and an F1-score of 88%, reflecting good accuracy in identifying severe ulcer stages.

Class	Precision	Recall	F1-Score	Support
Stage I	90%	94%	92%	93
Stage II	87%	87%	87%	94
Stage III	76%	77%	77%	83
Stage IV	91%	86%	88%	90

Table 18. ES-DDPM + DenseNet: Per-Class Metrics.

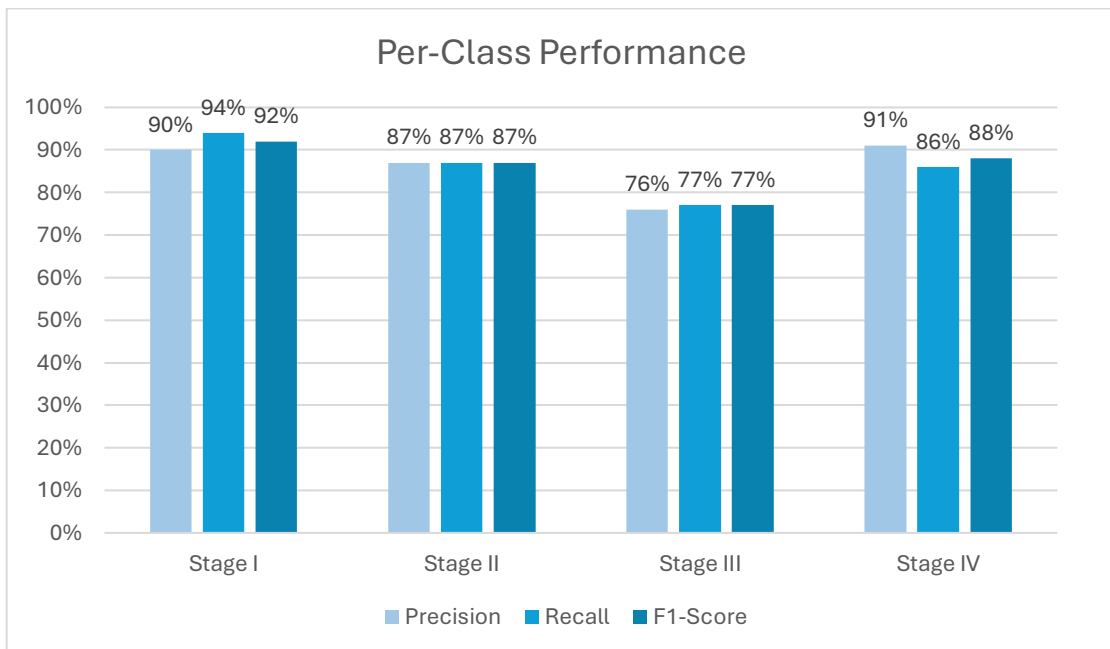


Figure 34. ES-DDPM + DenseNet Per-Class Performance Bar Chart.

The confusion matrix in Figure 35 provides further insight into the model's predictions. The diagonal elements (87, 82, 64, and 77) show that most samples were correctly classified across all stages, with misclassifications observed primarily between adjacent ulcer stages. For instance, 7 Stage II cases were misclassified as Stage I, and Stage III instances were confused with both Stage II (8 cases) and Stage IV (8 cases). This pattern suggests that the model struggles with mid-stage differentiation,

where texture and color characteristics often overlap. Notably, there is also a significant bleed from Stage IV, with 13 samples being misclassified as Stage III. Despite this, Stage IV showed high reliability, with 77 correctly classified images out of 90, reinforcing the model’s robustness in identifying advanced ulcers.

True Label	Stage I	87	4	2	0
	Stage II	7	82	5	0
	Stage III	3	8	64	8
	Stage IV	0	0	13	77
	Predicted Label	Stage I	Stage II	Stage III	Stage IV

Figure 35. ES-DDPM + DenseNet: Confusion Matrix.

Overall, the ES-DDPM + DenseNet model demonstrates strong and balanced classification performance with an accuracy of 86.11%, confirming the benefits of using ES-DDPM-generated data. The early-stopped diffusion process appears to have introduced valuable variability while maintaining sufficient image realism, allowing DenseNet to generalize better unseen data. Notably, the model’s performance improved compared to the non-augmented data setting (83.83%), suggesting that the augmented data successfully mitigated some of the limitations of the original small dataset. The consistent precision, recall, and F1-score across all metrics further indicate that ES-DDPM-enhanced data remains highly effective for training CNNs like DenseNet, particularly in improving robustness and reducing overfitting tendencies.

4.2.2 ES-DDPM + YOLOv11

The overall performance metrics of the ES-DDPM + YOLOv11 model indicate a strong classification ability, achieving an accuracy of 91%, with consistent macro and weighted averages for precision, recall, and F1-score all at around 90–91%. This reflects a well-balanced performance across all stages of pressure ulcers, suggesting that the YOLOv11 model benefitted from the enhanced image representations generated by the ES-DDPM process. The improvement in overall accuracy compared to its non-augmented data counterpart (88.44%) suggests that the synthetic variations introduced by ES-DDPM contributed to better generalization and robustness in identifying ulcer stages.

Macro Averages	
Precision	90.28%
Recall	90.31%
F1-score	90.27%
Weighted Averages	
Precision	90.51%
Recall	90.56%
F1-score	90.51%
Accuracy	
90.56%	

Table 19. ES-DDPM + YOLOv11: Core Performance.

The per-class metrics reveal strong results across all ulcer stages. I attained the highest performance, with 96% precision, 99% recall, and an F1-score of 97%, showing exceptional reliability in detecting early-stage ulcers. Stage II and Stage IV also achieved robust performance, both with F1-scores of 91%, indicating that the model successfully captured mid-to-severe ulcer characteristics. Although Stage III recorded the lowest F1-score (82%), it remains satisfactory, showing that YOLOv11 could still identify the more visually complex ulcer patterns, albeit with a few misclassifications. Overall, the per-class results demonstrate that ES-DDPM-generated data helped the model maintain accuracy and consistency across multiple stages.

Class	Precision	Recall	F1-Score	Support
Stage I	96%	99%	97%	93
Stage II	93%	89%	91%	94
Stage III	83%	81%	82%	83
Stage IV	89%	92%	91%	90

Table 20. ES-DDPM + YOLOv11s: Per-Class Metrics.

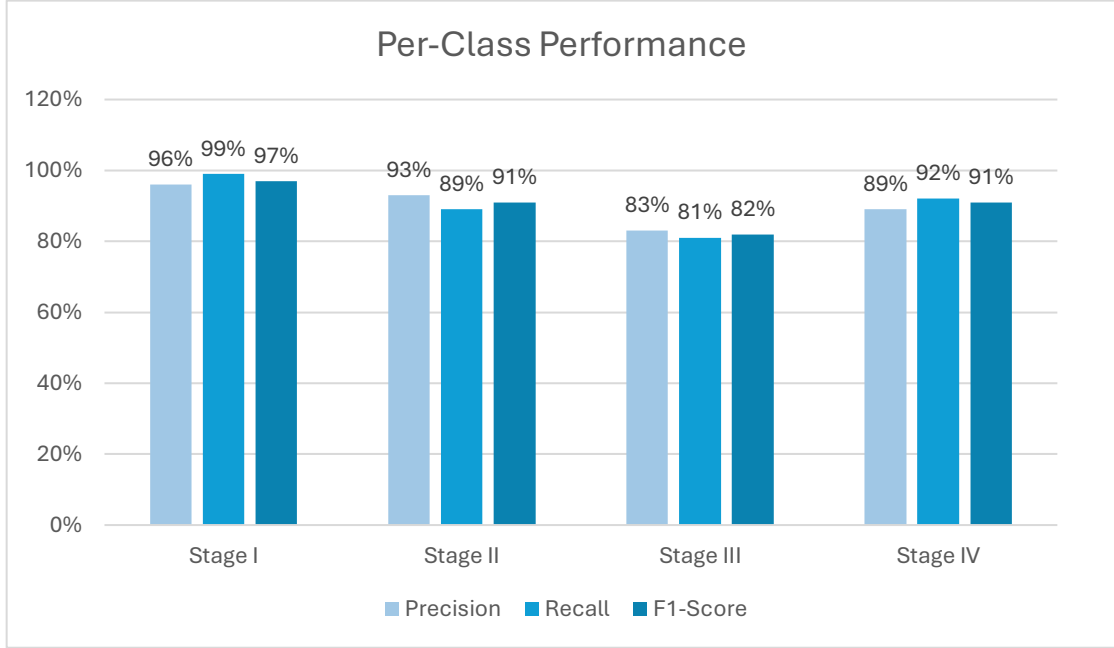


Figure 36. ES-DDPM + YOLOv11s Per-Class Performance Bar Chart.

The confusion matrix provides a detailed visualization of the model's predictions, with the strong diagonal elements (92, 84, 67, 83) indicating a high number of correctly classified samples across all stages. Stage I displayed exceptional accuracy, with 92 out of 93 images correctly identified. The most notable misclassifications occurred between adjacent stages of higher severity: 9 Stage II samples were mislabeled as Stage III, and similarly, 9 Stage III samples were mispredicted as Stage IV, reflecting some difficulty in distinguishing the exact boundaries as ulcer depth increases. Stage III proved the most challenging, with errors dispersed across all other classes. Despite these specific boundary overlaps, the high number of correct predictions in standard cases confirm YOLOv11's robust detection capability when trained with ES-DDPM-augmented data.

True Label	Stage I	Stage II	Stage III	Stage IV
	92	1	0	0
	0	84	9	1
	2	5	67	9
Predicted Label	Stage I	Stage II	Stage III	Stage IV
	2	0	5	83

Figure 37. ES-DDPM + DenseNet: Confusion Matrix.

The integration of ES-DDPM data into YOLOv11 training resulted in a notable performance increase, with an overall accuracy of 90.56%, which is an improvement over the non-augmented dataset baseline. This suggests that the generative augmentation process successfully enhanced the dataset's diversity and improved the model's feature recognition. The ES-DDPM approach helped YOLOv11 adapt to subtle texture variations and lighting conditions, improving its ability to generalize to unseen samples. However, some confusion between adjacent intermediate stages (particularly II and III) remains, which might be addressed by further fine-tuning or increasing the number of specifically targeted generated samples. In summary, ES-DDPM augmentation strengthened YOLOv11's capability to classify pressure ulcer stages, confirming the positive impact of generative modeling in improving CNN-based medical image classification.

4.2.3 ES-DDPM + CNN Models Comparative Analysis

When comparing the performance of ES-DDPM + DenseNet and ES-DDPM + YOLOv11, it is evident that YOLOv11 demonstrated superior classification results. The ES-DDPM + YOLOv11 model achieved an overall accuracy of 90.56%,

outperforming DenseNet's 86.11%, indicating that YOLOv11 benefited more significantly from the diverse representations generated by the ES-DDPM process. While both models showed balanced precision, recall, and F1-scores, YOLOv11 maintained higher consistency across all ulcer stages, specifically excelling in early (Stage I) and severe (Stage IV) classifications where it reached near-perfect recall. DenseNet, conversely, showed a more noticeable drop in performance for mid-stage ulcers, due to its high sensitivity to fine-grained texture patterns that may have been slightly altered by artifacts during the diffusion process. In contrast, YOLOv11's architecture appears to have adapted more effectively to these augmented variations. Overall, while both models benefited from generative augmentation, YOLOv11 exhibited a stronger response to the ES-DDPM-enhanced dataset, demonstrating improved generalization and robustness across all pressure ulcer stages.

4.3 StyleGAN2-ADA Data Setting

This section presents the results of applying StyleGAN2-ADA to the pressure ulcer dataset for synthetic data generation. The StyleGAN2-ADA approach leverages adaptive discriminator augmentation to stabilize training even with limited data, allowing the model to generate completely new, synthetic representations of each ulcer stage. The model attempts to capture essential textural and structural features from real data to produce realistic variations. This process aims to address data scarcity and class imbalance, providing a more voluminous and varied dataset for standard CNN-based classification. By generating these synthetic samples, the StyleGAN2-ADA model aims to fill gaps in the training data distribution, potentially leading to more robust generalization in succeeding experiments. However, as detailed in the subsequent analysis and shown in Figure 38, the model was unable to realistically generate images

with sufficient diversity across all stages, suffering from severe mode collapse and failing to meet the quality standards required for effective dataset augmentation.

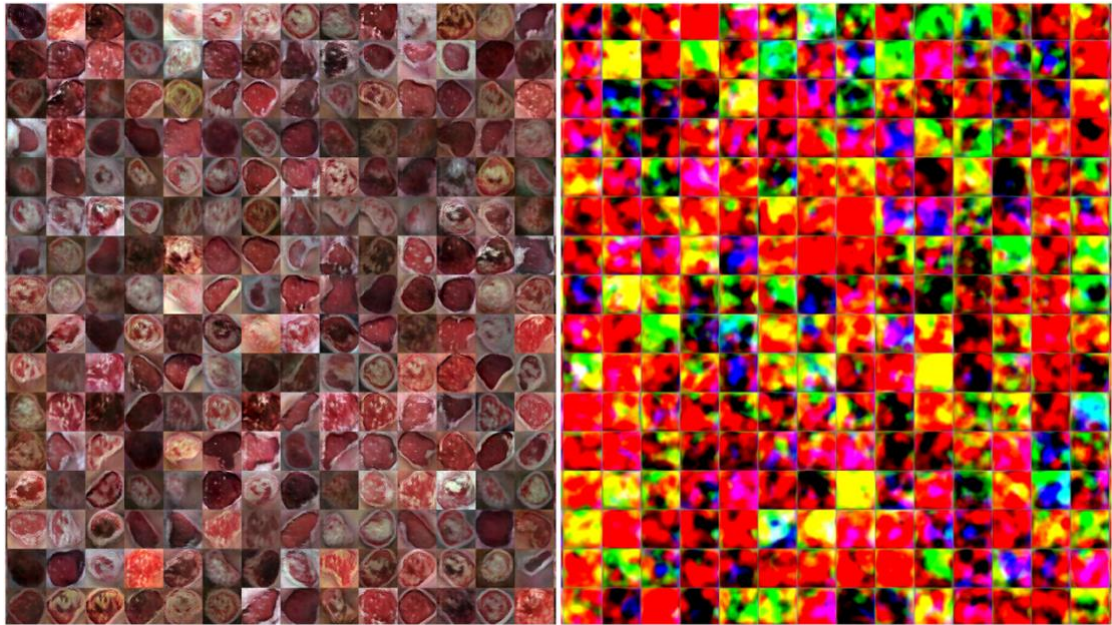


Figure 38. NON-AUGMENTED vs. StyleGAN2 Generated Samples

The generation of stage I pressure ulcers using StyleGAN2-ADA appeared to be the most challenging for the model (see Table 21). This stage yielded the highest Fréchet Inception Distance (FID) of 394.4817, a Kernel Inception Distance (KID) of 0.4136, a Precision and Recall of 0. This suggests a complete failure to generate images that realistically resemble genuine Stage 1 ulcers or capture any diversity from the training set. This deficient performance may be attributed to the visual subtlety of Stage 1 ulcers, which are characterized by intact skin with non-blanchable redness, features that might be difficult for the GAN to distinguish from noise or healthy skin without a sufficiently large and varied training set.

Stage I	
Metric	Score
Inception Score (IS)	2.066 ± 0.0098
Fréchet Inception Distance (FID)	394.4817
Kernel Inception Distance (KID)	0.4136
Precision	0
Recall	0

Table 21. StyleGAN2 Stage I Metrics.

For Stage II pressure ulcers, the model showed a marked improvement over Stage I, though challenges in diversity persisted (see Table 22). The Inception Score rose to approximately 3.78, and the FID decreased significantly to 157.2311, indicating generated images of higher visual quality that are statistically closer to the real data. The KID also improved to 0.1022. However, while a Precision score of 0.3136 suggests that a third of the generated images were realistic enough to fall within the manifold of real data, the extremely low Recall of 0.0020 indicates severe mode of collapse. This means the model is only capable of generating a very narrow variety of Stage 2 ulcer appearances, failing to represent the broader spectrum of wounds found in the actual dataset.

Stage II	
Metric	Score
Inception Score (IS)	3.782 ± 0.0298
Fréchet Inception Distance (FID)	157.2311
Kernel Inception Distance (KID)	0.1022
Precision	0.3136
Recall	0.0020

Table 22. StyleGAN2 Stage II Metrics.

The Stage III dataset yielded the most favorable comparative results among the four classes, a notable deviation from the NON-AUGMENTED and ES-DDPM settings where this intermediate stage was consistently difficult to discern. As shown in Table 23, the StyleGAN2-ADA model achieved the highest Inception Score of 3.967 (± 0.0287) and the lowest FID of 122.0696, suggesting that its distinct visual features, such as full-thickness skin loss, were easier for this specific generative architecture to learn. The KID further corroborates this with a low score of 0.0591. While the Precision (0.4450) represents the second-best fidelity among the stages, the Recall remains incredibly low at 0.0116. Although this is the highest Recall observed across all stages, it still points to a significant limitation in the model's ability to generate a diverse range of unique samples for this stage.

Stage III	
Metric	Score
Inception Score (IS)	3.967 ± 0.0287
Fréchet Inception Distance (FID)	122.0696
Kernel Inception Distance (KID)	0.0591
Precision	0.4450
Recall	0.0116

Table 23. StyleGAN2 Stage III Metrics.

The results for Stage IV present a case of high fidelity but absent diversity. The model achieved the highest Precision score of 0.6000, indicating that most of the generated images were assessed as realistic and comparable to real Stage 4 ulcers. However, the Recall score of 0.0000 reveals absolute mode of collapse; while the few images it produces are realistic, they look identical to each other, completely missing the varied presentations of Stage 4 ulcers (e.g., exposed bone, tendon, or muscle). The moderate FID of 255.8591 reflects this trade-off, where high individual image quality cannot fully compensate for the lack of distributional coverage.

Stage IV	
Metric	Score
Inception Score (IS)	3.204 ± 0.0223
Fréchet Inception Distance (FID)	255.8591
Kernel Inception Distance (KID)	0.1815
Precision	0.6000
Recall	0.0000

Table 24. StyleGAN2 Stage IV Metrics.

The application of StyleGAN2 to the pressure ulcer dataset revealed significant challenges in generating high-fidelity, diverse synthetic images across all four stages. While the model demonstrated some capability in producing realistic individual samples for Stages III and IV, it consistently suffered severe mode of collapse as shown by the near-zero Recall scores across all categories. This outcome confirms that standard GAN training, without robust augmentation techniques like ADA, is insufficient for small, highly variable medical datasets. Stage III proved to be the most

learnable class, due to its distinct visual features, while Stage I was the most difficult, with the model failing completely to capture its characteristics.

Given these results, the synthetic data generated by StyleGAN2 will not be integrated into the CNN classification models. The primary goal of using generative models in this context is to augment the training set with diverse examples to improve model generalization. However, in this specific experimental setup, the adaptive discriminator augmentation (ADA) was not utilized during training due to significant computational resource constraints. Moreover, the extreme lack of diversity means adding these images would introduce redundancy rather than latest information, potentially causing the classifiers to overfit the few repeated modes the GAN managed to learn. Furthermore, the inadequate quality of Stage I generations could actively harm model performance by introducing noisy or unrealistic data points. Therefore, relying on alternative generative approaches that are more stable on small datasets, like the ES-DDPM discussed previously, is deemed more suitable for this specific study.

4.4 Comparative Discussion

When comparing the overall performance of models trained on non-augmented data versus those trained on ES-DDPM-generated data, a clear trend of improvement emerges across both architectures. The models trained using standard non-augmented datasets established strong baselines, with DenseNet achieving 83.83% accuracy and YOLOv11 achieving 88.45%. The integration of ES-DDPM-enhanced datasets yielded universal performance gains: ES-DDPM + DenseNet improved to 86.11%, while ES-DDPM + YOLOv11 reached the highest overall accuracy in this study at 90.56%. These outcomes definitively demonstrate that the diffusion-based augmentation successfully mitigated limitations related to data scarcity and class imbalance, benefiting both

standard CNN and modern object detection architectures. This trend can be seen illustrated in Figure 39.

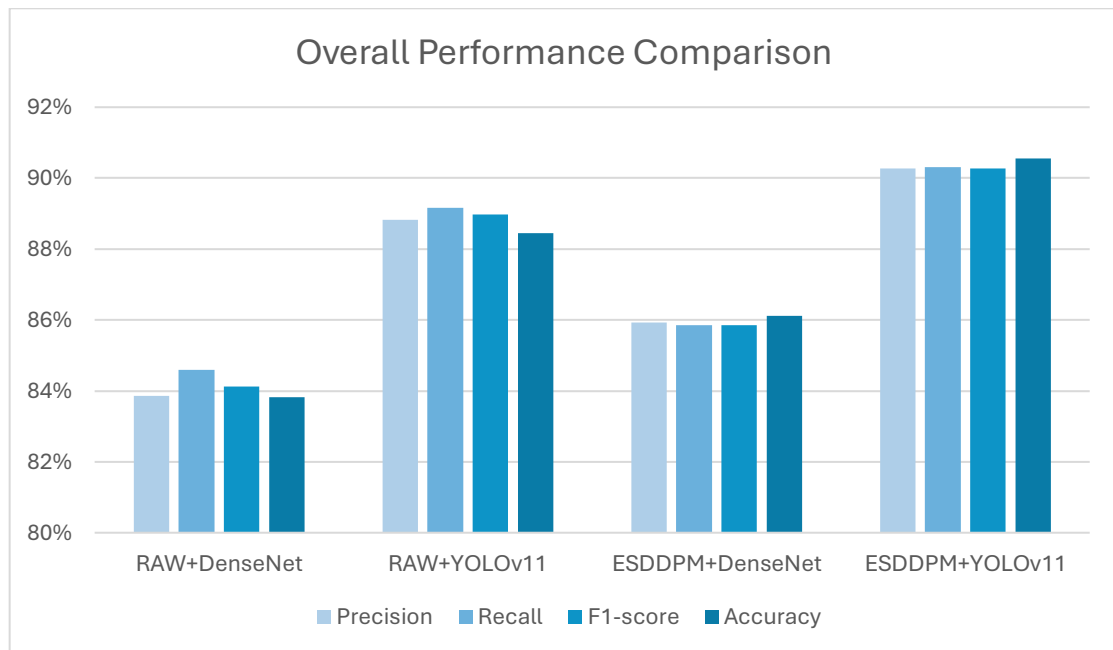


Figure 39. NON-AUGMENTED vs ES-DDPM Data Setting Overall Performance.

From a qualitative standpoint, the ES-DDPM process enhanced the dataset by introducing controlled variability while preserving essential wound characteristics. This increased diversity was particularly effective for YOLOv11, which utilized its spatially aware architecture to better generalize across unseen samples, especially in distinguishing the highly distinct visual patterns of early (Stage I) and severe (Stage IV) ulcers. While both models saw significant benefits from this augmentation, DenseNet's gain was +2.28% (from 83.83% to 86.11%), while YOLOv11's was +2.11% (from 88.45% to 90.56%). Although DenseNet's absolute gain was marginally larger, YOLOv11's ability to build on its already high baseline to achieve the study's top accuracy (90.56%) suggests its architecture was better suited to leverage the broader range of feature representations provided by the synthetic data.

Figure 40 reveals a consistent performance hierarchy across all models: Stage I achieved the highest scores, followed by Stage IV, then Stage II, with Stage III

consistently emerging as the weakest. This pattern aligns with the confusion matrix analysis, which highlights persistent challenges in distinguishing intermediate stages (Stages II and III) due to heavy overlap in visual symptoms such as wound depth and coloration. However, the ES-DDPM setting notably solidified performance at the extremes; for instance, the ES-DDPM + YOLOv11 model achieved near-perfect recognition of Stage I (92/93 correct). This indicates that while mid-stage ambiguity remains the primary challenge, the diffusion process successfully captured the distinct morphological features of early and late-stage ulcers, allowing the models to learn more robust decision boundaries where visual cues are most definitive.

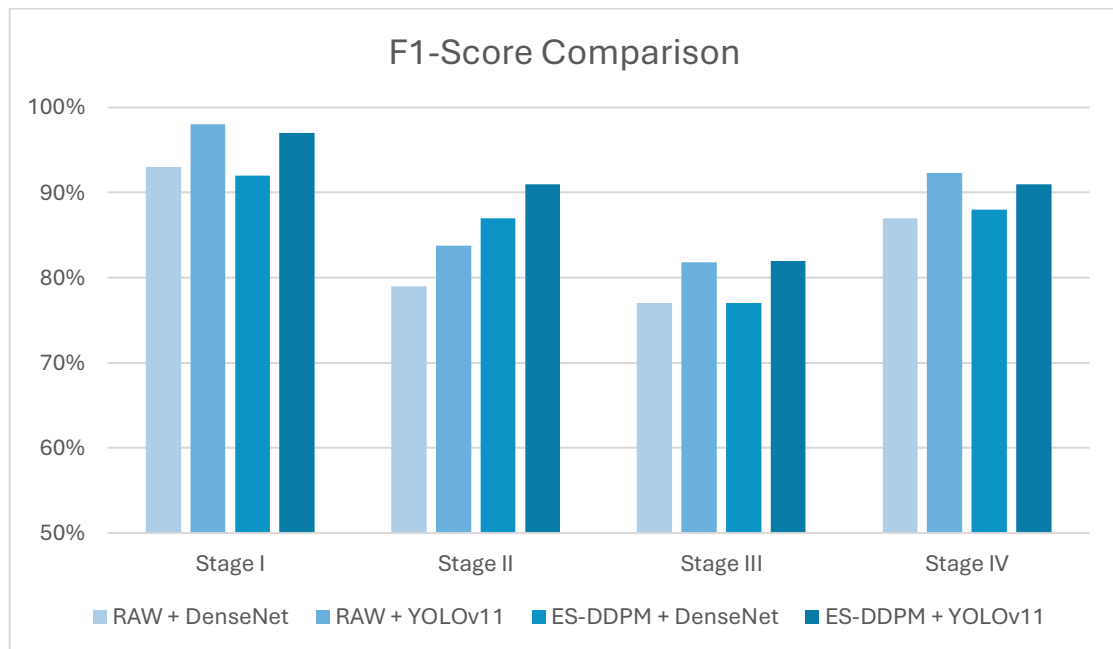


Figure 40. NON-AUGMENTED vs ES-DDPM F1-Score Comparison.

Table 25 shows a consolidated view of the performance metrics (Precision, Recall, F1-score) for each of the four pressure ulcer stages across the different experimental setups (NON-AUGMENTED + DenseNet, NON-AUGMENTED + YOLOv11, ES-DDPM + DenseNet, ES-DDPM + YOLOv11). The data highlights performance extremes: Stage I consistently achieved the highest scores, peaking with the study's best F1-score of 98% from the NON-AUGMENTED + YOLOv11 model.

Conversely, Stage III emerged as the most challenging class, having the lowest F1-scores, which bottomed out at 77% for both NON-AUGMENTED + DenseNet and ES-DDPM + DenseNet configurations. The table illustrates the targeted impact of the ES-DDPM augmentation, which yielded its most significant gains in Stage II, improving the F1-score from 79% to 87% for DenseNet and from 84% to 91% for YOLOv11. Notably, this augmentation did not resolve the ambiguity of Stage III, which saw no F1-score improvement for either architecture. The table also confirms the general superiority of the YOLOv11 architecture, as it outperformed DenseNet on the F1-score metric in all classes under both NON-AUGMENTED and ES-DDPM data settings.

Class	Metric	NON-AUGMENTED + DenseNet	NON-AUGMENTED + YOLOv11	ES-DDPM + DenseNet	ES-DDPM + YOLOv11
Stage I	Precision	89%	99%	90%	96%
	Recall	97%	97%	94%	99%
	F1-score	93%	98%	92%	97%
Stage II	Precision	82%	85%	87%	93%
	Recall	77%	83%	87%	89%
	F1-score	79%	84%	87%	91%
Stage III	Precision	80%	82%	76%	83%
	Recall	75%	82%	77%	81%
	F1-score	77%	82%	77%	82%
Stage IV	Precision	85%	90%	91%	89%
	Recall	89%	95%	86%	92%
	F1-score	87%	92%	88%	91%

Table 25. NON-AUGMENTED vs ES-DDPM Metrics per Class

Overall, the comparative analysis confirms that ES-DDPM functions not merely as a data augmentation technique, but as a critical enhancement strategy for medical image classification where data is limited. It consistently improved generalization and reduced overfitting tendencies across both tested architectures. Consequently, the combination of ES-DDPM synthetic data with the YOLOv11 model is established as the most effective approach in this study, offering a superior balance of precision, recall, and overall accuracy compared to reliance on non-augmented data alone.

Chapter 5

CONCLUSION AND RECOMMENDATIONS

This chapter presents the overall summary of the study, the key findings derived from the experimental results, and the conclusions drawn from the analysis. It also provides recommendations for future research directions and potential improvements to further enhance the performance and applicability of the proposed models in pressure ulcer classification.

5.1 Summary of Key Findings

This study explored the integration of generative models with convolutional neural networks (CNNs) to improve the classification of pressure ulcer stages (1–4). The dataset underwent multiple configurations: non-augmented images, ES-DDPM-enhanced images, StyleGAN-generated samples, and a hybrid DDPM-StyleGAN dataset. Each configuration was tested using two CNN architectures (DenseNet and YOLOv11) to determine the effect of generative augmentation on classification performance.

The findings found that among non-augmented data configurations, NON-AUGMENTED + YOLOv11 achieved the highest accuracy with 88.44%, surpassing NON-AUGMENTED + DenseNet at 83.83%. Upon integrating ES-DDPM, both models saw performance gains: YOLOv11 improved to 90.56%, demonstrating superior generalization, while DenseNet increased to 86.11%. This indicates that generative augmentation successfully enhanced data diversity and reduced overfitting for both architectures, though the modern YOLOv11 model leveraged these synthetic variations more effectively to reach peak classification reliability.

5.2.1 The Baseline Problem: Architectural Superiority and Mid-Stage Ambiguity

The initial experimental phase, which utilized the original un-augmented (NON-AUGMENTED) dataset, established two critical facts.

First, it confirmed a clear difference in architectural suitability for this task. The modern YOLOv11 architecture (88.44% accuracy) demonstrably outperformed the DenseNet model (83.83% accuracy) on the non-augmented dataset, establishing it as the superior baseline.

Second, and more importantly, this phase identified the central research challenge. Both models, regardless of their overall accuracy, excelled at identifying the visually distinct extremes of the condition: Stage I (98.04% F1-score for YOLOv11) and Stage IV (92.31% F1-score for YOLOv11). However, both models "struggled with the intermediate stages." The confusion matrices for both DenseNet and YOLOv11 revealed that the "most prominent confusion occurring between Stage II and Stage III." This "mid-stage boundary ambiguity" became the primary problem to be solved by generative data augmentation.

5.2.2 The Generative Model Experiments: A Tale of Failure and Success

The study's core hypothesis was that generative models could create novel, realistic training data to help classifiers resolve the mid-stage ambiguity. This led to two starkly different outcomes.

1. The Generative Failure of StyleGAN2-ADA. A parallel experimental track attempting to use StyleGAN2-ADA for data generation resulted in a critical "null result": the model failed to produce a usable dataset. This was not a subjective assessment but a quantitative one; the model experienced "severe mode of collapse" across all classes. This was most evident in the Recall metric, which measures diversity.

For Stage IV, the Recall was 0.0000, and for Stage I, it was 0, with a similar catastrophic 0.0020 for Stage II.

This "extreme lack of diversity" meant the generated images would "introduce redundancy rather than new information." A critical methodological decision was therefore made: "the synthetic data generated by StyleGAN2 will not be integrated into the CNN classification models." This finding illustrates a significant limitation of this specific GAN architecture, which, due to "significant computational resource constraints," was unable to handle the small, high-variance medical dataset.

2. The Augmentation Success of ES-DDPM. In direct contrast, the Early-Stopped Denoising Diffusion Probabilistic Model (ES-DDPM) proved to be a successful augmentation strategy. This approach "successfully mitigated limitations related to data scarcity" by introducing controlled variability. The impact was a clear and universal performance uplift for both classifiers. The ES-DDPM data pushed DenseNet's accuracy from 83.83% to 86.11% (a +2.28% gain). It had a similarly positive effect on the superior YOLOv11 architecture, raising its accuracy from 88.45% to 90.56% (a +2.11% gain). This result established the ES-DDPM + YOLOv11 combination as the champion model of this study, successfully breaking the 90% accuracy threshold.

5.2 Conclusion

This thesis embarked on a systematic investigation to advance the automated classification of pressure ulcer stages using deep learning. The study first established a robust performance baseline with non-augmented data, confirming the architectural superiority of YOLOv11 (88.44% accuracy) over DenseNet (83.83% accuracy). The primary contribution of this work was the critical evaluation of two advanced generative augmentation strategies. This evaluation demonstrated the profound success of Early-

Stopped Denoising Diffusion Probabilistic Models (ES-DDPM), which successfully enhanced dataset diversity and elevated the YOLOv11 model to a state-of-the-art 90.56% accuracy. Concurrently, this thesis provides a significant "null result" by demonstrating the total generative failure of StyleGAN2-ADA due to severe mode collapse, a cautionary finding for researchers applying standard GANs to small, high-variance medical datasets.

The most significant insight from this work is the identification of the primary *persistent limitation* of such systems: even with successful augmentation, the intrinsic visual ambiguity between Stage II and Stage III pressure ulcers remains a formidable challenge. This thesis concludes that the ES-DDPM + YOLOv11 model is the most effective and promising approach for deployment identified in this study. However, it recommends a "human-in-the-loop" strategy for clinical integration, a system that specifically leverages human expertise to resolve the ambiguous intermediate-stage classifications that augmentation can ameliorate but not yet fully resolve. This work successfully charts a path forward, highlighting both the most promising tool and the most critical challenge for the next generation of automated wound care.

5.3 Recommendations

For future studies, the following recommendations are proposed:

1. Hybrid Generative Techniques: Explore the combination of ES-DDPM and StyleGAN to produce higher-quality and more diverse synthetic data.
2. Larger Dataset and Clinical Validation: Apply the models to larger and clinically validated wound datasets to evaluate real-world performance.

3. Explainability and Visualization: Incorporate explainable AI (XAI) tools such as Grad-CAM or saliency mapping to visualize which wound features influence model predictions.
4. Integration into Clinical Systems: Develop an end-to-end framework for real-time wound assessment using the optimized YOLOv11 model.
5. Comparison with Other Architectures: Evaluate transformer-based models (e.g., Vision Transformers or ConvNeXt) to further benchmark performance against CNNs.

In summary, this research confirms that diffusion-based generative modeling can improve the adaptability and robustness of CNNs, paving the way for more reliable automated pressure ulcer classification systems in the medical field.

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